

Loving or Fighting Inflation: Optimal Policy in High- and Low-Debt Economies*

Diego Rodrigues[†]

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Abstract

The rise in public debt after the COVID-19 pandemic and the energy crisis has renewed debate over how fiscal and monetary policies should coordinate in highly indebted economies. We study optimal fiscal and monetary policy in high- and low-debt environments, using the United Kingdom in 2007 and 2021 as benchmarks. Within a heterogeneous-agent Ramsey framework with incomplete markets, we analyze how taxation, public debt, and inflation adjust following fiscal and productivity shocks. Higher public debt limits fiscal space, requiring sharper and more front-loaded tax adjustments and shifting the burden of stabilization toward monetary policy. When fiscal tools are constrained, moderate inflation becomes an endogenous and welfare-improving adjustment mechanism, reducing the real value of debt and smoothing consumption.

Keywords: Heterogeneous agents, Ramsey policies, Monetary Policy, Fiscal Policy, Inflation, Public Debt.

JEL codes: E61, E62, E63, E52, D52, H21.

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[†]Corresponding author: Université du Québec à Montréal (UQAM), Department of Economics, 320 rue Ste-Catherine Est, Montréal (Québec), Canada H2X 1L7. Phone: +1 (514) 987-3000. E-mail: de_sousa_rodrigues.diego@uqam.ca.

1 Introduction

The evolving economic landscape, exacerbated by events such as COVID-19 and recent energy crises, underscores the critical role of government intervention in cushioning economies against shocks and fostering recovery. As countries contemplate further public spending and debt issuance to address these challenges, significant shifts in fiscal structures have become evident, as exemplified by the case of the United Kingdom.

Between 2007 and 2021, the United Kingdom experienced major transformations in both the size of the state and income inequality. In 2007, the public debt-to-GDP ratio stood at 43.1%, and income inequality was moderate, with a post-tax Gini coefficient of 38.6. By 2021, however, the fiscal landscape had changed dramatically: the debt-to-GDP ratio more than doubled to 101%, signaling an expanded role for the state in the economy. Over the same period, income inequality declined substantially, with the post-tax Gini coefficient falling from 38.6 to 29.9 (Table 1). Meanwhile, inflation has surged in recent years, posing renewed challenges for monetary authorities (Figure 1). These developments raise a central question: how should fiscal and monetary authorities coordinate in high-debt environments to absorb shocks and maintain macroeconomic stability?

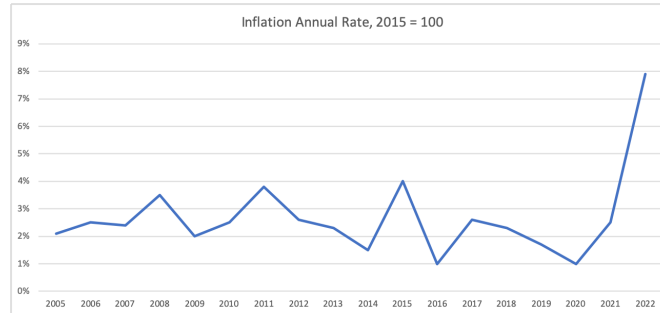


Figure 1: Annual Consumer Price Index (CPI) inflation rate, all items (2015 = 100). Source: Office for National Statistics.

We address this question using a Ramsey framework with heterogeneous agents and incomplete markets, considering two distinct scenarios. One corresponds to a low debt-to-GDP ratio, calibrated to the UK in 2007; the other reflects the high debt ratio observed in 2021. By comparing these two regimes, we assess whether recent changes in fiscal policy environments represent a shift in optimal responses to shocks and whether high-debt economies may find it optimal to tolerate higher inflation as an adjustment mechanism.

Year / Economy	Debt-to-GDP (B/Y)	Gini (Pre-Tax)	Gini (Post-Tax)
United Kingdom (2007)	43.1	53.5	38.6
United Kingdom (2021)	101.0	50.2	29.9

Table 1: Public Debt and Income Inequality in the United Kingdom, 2007 and 2021. Source: Office for National Statistics.

To analyze optimal dynamic responses, we follow a structured approach. First, we construct calibrated models reflecting the macroeconomic conditions of the UK in 2007 and 2021. The year 2007 precedes the global financial crisis and the COVID-19 pandemic—two pivotal events that reshaped fiscal structures—while 2021 represents the post-crisis, high-debt environment. Extended periods of elevated debt may indicate a new fiscal “steady state,” making it crucial to understand how optimal policies adjust under such conditions.

Our model includes capital accumulation, progressive labor taxation, capital taxation, public debt, and inflation, with incomplete markets for idiosyncratic risk as the only friction. Incorporating monetary policy allows us to observe how inflation and taxes interact in managing debt dynamics—highlighting the role of inflation as a policy tool, particularly when fiscal instruments are constrained.

We assume that observed fiscal outcomes and income distributions for each year reflect the optimal choices of a benevolent planner who maximizes intertemporal welfare in a heterogeneous-agent economy. The planner possesses a Social Welfare Function (SWF) that internalizes the distortions and general-equilibrium effects of all fiscal and monetary instruments.

Given our focus on the joint dynamics of fiscal and monetary policy, we begin by estimating an SWF consistent with the observed debt and inequality measures for the UK. Following the inverse-optimum taxation approach of [Heathcote & Tsujiyama \(2021\)](#) and [Le Grand et al. \(2025\)](#), we infer the implicit social weights that rationalize observed fiscal and redistributive outcomes as optimal steady-state choices. This estimation provides the welfare foundation for our dynamic analysis.

Analyzing a model with multiple fiscal and monetary instruments poses computational challenges. To address these, we employ a sequential representation of heterogeneous-agent models and apply the truncation procedure of [Le Grand & Ragot \(2022a\)](#) to obtain a finite-dimensional state space. We then use the Lagrangian approach of [Marcet & Marimon \(2019\)](#) to derive the first-order conditions of the Ramsey problem with commitment.

Once the model approximates the steady states of the UK in 2007 and 2021, we study the optimal responses to public spending and productivity shocks of varying persistence. By introducing both demand and supply shocks, we observe how key variables evolve around these calibrated steady states. The results reveal distinct optimal strategies depending on

shock persistence, especially in the management of public debt and inflation. In the absence of monetary policy, fiscal adjustments—through debt and taxes—serve as the main stabilization instruments, and their intensity depends on the persistence of the disturbance. For instance, in response to a positive government spending shock, the decline in capital prompts the planner to consider increasing public debt to facilitate consumption smoothing. However, the optimal response varies with shock persistence. In scenarios of low shock persistence, the planner opts to increase public debt to counteract the fall in capital. Conversely, in high-persistence scenarios, a sustained capital decline makes increasing public debt more costly, leading the planner to anticipate future tax hikes instead.

Comparing 2007 and 2021, higher initial debt levels in 2021 require sharper and more immediate tax increases to stabilize public debt. Fiscal adjustments are thus more front-loaded and contractionary, reflecting reduced fiscal space relative to 2007.

When monetary instruments are introduced, the adjustment dynamics change substantially. The planner may increase public debt even under persistent shocks, as higher interest rates help stabilize inflation and mitigate capital losses. Public debt thus serves as a short-run buffer, complementing monetary tightening. These results contrast with [LeGrand & Ragot \(2025\)](#), who emphasize that the optimal debt path depends on shock persistence. In our framework, when both fiscal and monetary policies are active, inflation remains close to target through coordinated interest rate and fiscal adjustments.

We extend the analysis in two directions. First, we impose fiscal feedback rules that restrict tax flexibility and show that inflation becomes an implicit stabilizer when fiscal tools are rigid. Second, we introduce a monetary rule—representing an independent central bank—and show that when fiscal flexibility is absent, temporary inflation can be welfare-improving by alleviating debt burdens and supporting demand. The persistence of the shock determines the magnitude of the inflation response, which is considerably larger in 2021 than in 2007.

Overall, the analysis yields three main insights. (i) High debt levels fundamentally alter optimal policy responses by constraining fiscal space and shifting adjustment toward inflation. (ii) Inflation acts as an endogenous fiscal instrument—an implicit tax on nominal assets—when explicit taxation is politically or institutionally constrained. (iii) Moderate, temporary inflation can be welfare-improving in high-debt economies, provided it remains well-anchored and consistent with long-run fiscal sustainability. Together, these results highlight the importance of fiscal–monetary coordination in an era of persistently high public debt.

Related literature. Our paper contributes to different streams of the literature. We contribute first to the literature that seeks to infer social preferences from observed fiscal systems. This strand, known as the inverse optimal taxation problem, estimates the Social

Welfare Function implied by actual tax and transfer schedules, as demonstrated by [Bargain & Keane \(2010\)](#), [Bourguignon & Amadeo \(2015\)](#), [Chang et al. \(2018\)](#), [Heathcote & Tsujiyama \(2021\)](#), and more recently [Le Grand et al. \(2025\)](#). Additional foundational works in this area include [Saez \(2001\)](#) and [Saez & Stantcheva \(2016\)](#), who provide frameworks for deriving welfare weights from microdata and optimal taxation theory. [Lockwood & Weinzierl \(2015\)](#) also explore how heterogeneity in individual preferences influences optimal redistribution. This literature connects closely to the estimation of country-specific welfare weights, as in [Piketty & Saez \(2013\)](#), and to the broader understanding of how fiscal policy reflects normative trade-offs between equity and efficiency. Our paper apply this framework by estimating welfare weights consistent with the U.K. fiscal system and then embedding them into a dynamic heterogeneous-agent Ramsey model to study how optimal tax, debt, and inflation policies evolve under different debt regimes. In doing so, we use the methodology developed by [Le Grand et al. \(2025\)](#) in a new context—applying it to a different country and contrasting fiscal environments.

Second, our paper contributes to the literature on optimal fiscal policy with heterogeneity and incomplete markets. Classical studies such as [Werning \(2007\)](#) and [Bassetto \(2014\)](#) analyze intertemporal tax-smoothing under commitment, while [Bhandari et al. \(2017\)](#) and [Farhi & Werning \(2013\)](#) incorporate idiosyncratic risk and borrowing constraints into the Ramsey framework. While existing studies have explored the effects of fiscal experiments in such frameworks, including those by [Heathcote \(2005\)](#) and [Kaplan & Violante \(2014\)](#), our work extends this by considering equilibrium multiplicity in an economy with capital, as evidenced by [Dávila et al. \(2012\)](#). Our work is also related to the foundational contributions of [Aiyagari \(1995\)](#) and [Aiyagari & McGrattan \(1998\)](#), who introduced incomplete markets into optimal public debt analysis, and to [Chari & Kehoe \(1999\)](#) and [Barro \(1979\)](#), who studied debt and taxation in representative-agent settings. Moreover, we add to the literature by solving a Ramsey problem to determine optimal tax systems, following the approach of [Dyrda & Pedroni \(2022\)](#) and [Açikgöz et al. \(2022\)](#). We build on this literature by comparing two fiscal regimes calibrated to the U.K. economy in 2007 and 2021, demonstrating how higher initial debt compresses fiscal space and alters optimal trade-offs between taxes, debt, and inflation.

Third, our work intersects with the literature on optimal monetary policy under heterogeneity and limited fiscal space. The redistributive effects of monetary policy in incomplete markets have been extensively studied by [Kaplan et al. \(2018\)](#), [Auclert \(2019\)](#), and [Gornemann et al. \(2016\)](#), with related discussions in [LeGrand et al. \(2025\)](#) and [Hagedorn et al. \(2019\)](#). Other references in the strand of optimal monetary policy with incomplete markets can be seen in [Acharya et al. \(2019\)](#), [Bhandari et al. \(2021\)](#), and [Nuno & Thomas \(2019\)](#). Earlier contributions such as [Leeper \(1991\)](#), [Schmitt-Grohé & Uribe \(2004\)](#), and [Woodford & Walsh \(2005\)](#) analyze the interaction between fiscal and monetary authorities under commitment and price stickiness. Recent work on the fiscal theory of the price level by [Cochrane \(2023\)](#) and [Sims \(2013\)](#) has further highlighted the importance of coordina-

tion between fiscal and monetary policy. Our paper contributes to this debate by explicitly comparing optimal policy responses in high- and low-debt regimes, showing that when fiscal instruments are constrained, the planner optimally tolerates moderate inflation as a state-contingent fiscal tool. Notably, we innovate by introducing fiscal tools and comparing their effects in the absence of fiscal mechanisms. Our work builds upon the findings of [LeGrand & Ragot \(2025\)](#) by incorporating monetary policy considerations and demonstrating that in certain circumstances, it may be optimal to increase debt even in scenarios of high persistent shocks. We also show that the presence of monetary policy alters the relationship between shock persistence and optimal debt paths identified by [LeGrand & Ragot \(2025\)](#).

Finally, our work connects to studies on the redistributive and debt-management role of inflation. [Doepke & Schneider \(2006\)](#) show how inflation redistributes nominal wealth across agents, while [Erosa & Ventura \(2002\)](#) and [Albanesi \(2007\)](#) analyze its regressive effects through consumption taxes. At the macro level, [Hilscher et al. \(2022\)](#) and [Blanchard \(2019\)](#) discuss inflation as a mechanism for reducing real debt burdens in high-debt economies. In the context of optimal policy, recent studies by [Nuño & Thomas \(2022\)](#), [McLeay & Tenreyro \(2020\)](#), [Blanco \(2021\)](#), [McKay & Wolf \(2022\)](#), and [Bilbiie & Ragot \(2021\)](#) have characterized the welfare and stabilization trade-offs of nonzero inflation in heterogeneous-agent models. Our findings complement this literature by showing that inflation naturally emerges as an endogenous fiscal adjustment margin in high-debt environments, allowing temporary deviations from zero inflation to smooth debt and consumption paths.

Overall, our contribution is threefold. First, we estimate social welfare weights consistent with observed inequality and fiscal incidence for two distinct U.K. regimes (2007 and 2021). Second, we integrate these empirical welfare measures into a dynamic heterogeneous-agent Ramsey framework to quantify how debt levels shape optimal policy trade-offs. Third, we show that when fiscal tools are restricted, inflation optimally assumes a stabilizing role, redistributing resources and mitigating the fiscal costs of adjustment in high-debt economies.

The structure of our paper is as follows: Section 2 presents the Model, Section 3 covers the Ramsey Problem and the conditions for obtaining the allocation under Optimal Fiscal and Monetary Policy, and Section 4 outlines the quantitative investigation and numerical approach to solving this class of problem. Finally, Section 5 concludes the paper.

2 The model

Time is discrete and indexed by $t \geq 0$. The economy is populated by a continuum of ex-ante identical agents. The population, normalized to size one, is distributed over a segment J according to a non-atomic measure ℓ , such that $J(\ell) = 1$. Following [Green \(1994\)](#), we assume that the law of large numbers holds.

2.1 Risk structure

At the beginning of each period, agents face an uninsurable idiosyncratic labor productivity shock y , which can take Y discrete values in the set $\mathcal{Y} \subset \mathbb{R}_+$. The productivity process follows a first-order Markov chain with transition matrix $(\pi_{yy'})_{y,y'}$.

In the stationary distribution, the fraction of agents with productivity level y is denoted by S_y and satisfies $S_y = \sum_{\tilde{y} \in \mathcal{Y}} \pi_{\tilde{y}y} S_{\tilde{y}}$, $\forall \tilde{y} \in \mathcal{Y}$. Normalizing average productivity to one implies $\sum_{y \in \mathcal{Y}} S_y y = 1$.

For each agent i , let $y_i^t = \{y_{i,0}, \dots, y_{i,t}\} \in \mathcal{Y}^{t+1}$ denote the history of idiosyncratic shocks up to period t , where $y_{i,\tau}$ is the productivity realization in period τ . The measure of idiosyncratic histories at date t is denoted by θ_t and can be derived from the transition probabilities. In particular, $\theta_t(y^t)$ represents the share of agents with history y^t at time t .

In addition to idiosyncratic risk, the economy is subject to an aggregate productivity shock, denoted by $(z_t)_{t \geq 0}$, which follows an AR(1) process. Aggregate productivity is given by $Z_t = Z_0 e^{z_t}$, and the history of aggregate shocks up to period t is $z^t = \{z_0, \dots, z_t\} \in \mathbb{R}^{t+1}$.

Each agent i chooses labor supply $l_{i,t}$ and earns a before-tax wage rate $\tilde{w}_t$. Hence, total before-tax labor income in period t is given by $\tilde{w}_t y_{i,t} l_{i,t}$.

2.2 Preferences

Agents derive utility from consumption and disutility from labor. Each agent i values sequences of consumption $(c_{i,t})_{t \geq 0}$ and labor $(l_{i,t})_{t \geq 0}$ according to a time-separable utility function $\sum_{t=0}^{\infty} \beta^t U(c_{i,t}, l_{i,t})$, where $\beta \in (0, 1)$ is the discount factor. The period utility function $U(c_{i,t}, l_{i,t})$ is assumed to be separable in consumption and labor:

$$U(c_{i,t}, l_{i,t}) = u(c_{i,t}) - v(l_{i,t}). \quad (1)$$

The function $u : \mathbb{R}_+ \rightarrow \mathbb{R}$ represents the utility from consumption and is twice continuously differentiable, strictly increasing, and strictly concave, with $u'(0) = \infty$. The function $v : \mathbb{R}_+ \rightarrow \mathbb{R}$ represents the disutility from labor and is twice continuously differentiable, strictly increasing, and strictly convex, with $v'(0) = 0$.

2.3 Production

There is a continuum of monopolistically competitive firms indexed by $j \in [0, 1]$, each producing a differentiated variety $Y_{j,t}$. These intermediate goods are aggregated into a final composite good according to a constant-elasticity-of-substitution (CES) aggregator with elasticity

of substitution ε :

$$Y_t = \left[\int_0^1 Y_{j,t}^{\frac{\varepsilon-1}{\varepsilon}} dj \right]^{\frac{\varepsilon}{\varepsilon-1}}.$$

Profit maximization by the final-good producer yields the standard demand function for each variety and the associated aggregate price index:

$$Y_{j,t} = \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t, \quad \text{and} \quad P_t = \left[\int_0^1 P_{j,t}^{1-\varepsilon} dj \right]^{\frac{1}{1-\varepsilon}}.$$

Intermediate firms produce using a Cobb–Douglas technology,

$$Y_{j,t} = Z_t k_{j,t}^\alpha l_{j,t}^{1-\alpha},$$

where Z_t is the aggregate productivity level, $k_{j,t}$ denotes the firm's capital input, and $l_{j,t}$ denotes labor. In equilibrium, the output of each firm equals the demand for its variety from the final-good producer, with intermediate goods sold at the relative price $P_{j,t}/P_t$.

Let $\tilde{w}_t$ denote the real before-tax wage, and let $\tilde{r}_t^k$ denote the real before-tax rental rate of capital (net of depreciation $\delta > 0$). The firm's cost-minimization problem yields:

$$\tilde{r}_t^k + \delta = \xi_{j,t} \alpha \frac{Y_{j,t}}{k_{j,t}} \quad \text{and} \quad \tilde{w}_t = \xi_{j,t} (1 - \alpha) \frac{Y_{j,t}}{l_{j,t}}, \quad (2)$$

where $\xi_{j,t}$ is the Lagrange multiplier associated with the production constraint. Optimality implies a common value ξ_t across firms:

$$\xi_t = \frac{1}{Z_t} \left(\frac{\tilde{r}_t^k + \delta}{\alpha} \right)^\alpha \left(\frac{\tilde{w}_t}{1 - \alpha} \right)^{1-\alpha}. \quad (3)$$

Aggregating the factor-price conditions in (2) across firms and using the production function yields expressions for aggregate capital and labor demand:

$$K_{t-1} = \frac{1}{Z_t} \left(\frac{\tilde{r}_t^k + \delta}{\alpha} \right)^{\alpha-1} \left(\frac{\tilde{w}_t}{1 - \alpha} \right)^{1-\alpha} Y_t, \quad L_t = \frac{1}{Z_t} \left(\frac{\tilde{r}_t^k + \delta}{\alpha} \right)^\alpha \left(\frac{\tilde{w}_t}{1 - \alpha} \right)^{-\alpha} Y_t. \quad (4)$$

Aggregate output can therefore be expressed as:

$$Y_t = Z_t K_{t-1}^\alpha L_t^{1-\alpha} = \frac{(\tilde{r}_t^k + \delta) K_{t-1} + \tilde{w}_t L_t}{\xi_t},$$

and factor ratios satisfy:

$$\frac{K_{t-1}}{L_t} = \left(\frac{\alpha}{1 - \alpha} \right) \left(\frac{\tilde{w}_t}{\tilde{r}_t^k + \delta} \right). \quad (5)$$

In a perfectly competitive setup, $\xi_{j,t} = \xi_t = 1$, yielding the standard expressions for factor

prices: $\tilde{r}_t^k + \delta = \alpha Z_t \left(\frac{K_{t-1}}{L_t} \right)^{\alpha-1}$ and $\tilde{w}_t = (1 - \alpha) Z_t \left(\frac{K_{t-1}}{L_t} \right)^\alpha$.

In addition to the production costs associated with acquiring inputs from households—capital and labor as in (2)—we assume that firms face a quadratic price adjustment cost à la Rotemberg (1982) when setting their prices. Following the literature, this adjustment cost is proportional to the magnitude of the firm's relative price change and is given by $\frac{\psi}{2} \left(\frac{P_{j,t}}{P_{j,t-1}} - 1 \right)^2 Y_t$, where $\psi \geq 0$ measures the degree of price rigidity. Letting P_t denote the aggregate price index of the economy, we can thus express the real profit of firm j at date t , denoted by $D_{j,t}$, as:

$$D_{j,t} = \frac{P_{j,t}}{P_t} Y_{j,t} - \frac{(1 - \tau_t^D)}{Z_t} \left(\frac{\tilde{r}_t^k + \delta}{\alpha} \right)^\alpha \left(\frac{\tilde{w}_t}{1 - \alpha} \right)^{1-\alpha} Y_{j,t} - \frac{\psi}{2} \left(\frac{P_{j,t}}{P_{j,t-1}} - 1 \right)^2 Y_t - t_t^D,$$

where t_t^D is a lump-sum tax used to finance the production subsidy τ_t^D . Substituting the equilibrium relationships yields:

$$D_{j,t} = \left(\frac{P_{j,t}}{P_t} - \xi_t(1 - \tau_t^D) \right) \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} Y_t - \frac{\psi}{2} \left(\frac{P_{j,t}}{P_{j,t-1}} - 1 \right)^2 Y_t - t_t^D. \quad (6)$$

Firm j chooses its price path $(P_{j,t})_{t \geq 0}$ to maximize the expected discounted value of profits:

$$\max_{\{P_{j,t}\}_{t \geq 0}} \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \frac{M_t}{M_0} D_{j,t} \right],$$

where β is the discount factor (as households own firms) and $\frac{M_t}{M_0}$ is the pricing kernel.

The corresponding first-order condition is:

$$\begin{aligned} & \left[(1 - \varepsilon) + \varepsilon \xi_t (1 - \tau_t^D) \frac{P_t}{P_{j,t}} \right] \left(\frac{P_{j,t}}{P_t} \right)^{-\varepsilon} \frac{Y_t}{P_t} - \psi (\Pi_t - 1) \Pi_t \frac{Y_t}{P_{j,t}} \\ & + \beta \psi \mathbb{E}_t \left[\frac{M_{t+1}}{M_t} (\Pi_{t+1} - 1) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \right] \frac{Y_t}{P_{j,t}} = 0, \end{aligned}$$

where $\Pi_t \equiv P_{j,t}/P_{j,t-1}$ denotes gross inflation.

Setting $\tau_t^D = 1/\varepsilon$ ensures an efficient steady state. In a symmetric equilibrium, $P_{j,t} = P_t$ for all j , yielding:

$$(1 - \varepsilon) + \varepsilon \xi_t \left(\frac{\varepsilon - 1}{\varepsilon} \right) - \psi (\Pi_t - 1) \Pi_t + \beta \psi \mathbb{E}_t \left[(\Pi_{t+1} - 1) \Pi_{t+1} \frac{Y_{t+1}}{Y_t} \frac{M_{t+1}}{M_t} \right] = 0.$$

This leads to the New Keynesian Phillips curve:

$$\Pi_t (\Pi_t - 1) = \frac{\varepsilon - 1}{\psi} (\xi_t - 1) + \beta \mathbb{E}_t \left[\Pi_{t+1} (\Pi_{t+1} - 1) \frac{Y_{t+1}}{Y_t} \frac{M_{t+1}}{M_t} \right]. \quad (7)$$

Finally, in equilibrium the firm's real profits are identical across firms and given by:

$$D_t = \left(1 - \xi_t - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t. \quad (8)$$

2.4 Assets

Agents have access to two types of assets. The first is nominal public debt, with aggregate supply denoted by B_t , determined by the government. This asset pays a predetermined nominal gross interest rate before taxes. The nominal rate between periods $t - 1$ and t is known and denoted by $\tilde{R}_{t-1}^n$. The corresponding real interest rate is $\tilde{R}_{t-1}^n/\Pi_t$, where Π_t denotes the gross inflation rate.

The second asset is equity (capital shares), which yields a before-tax return $\tilde{r}_t^k$, as described above. The aggregate capital stock in the economy is denoted by K_t .

Public debt and capital are assumed to be perfect substitutes and to yield the same pre-tax return $\tilde{r}_t$. This assumption is analogous to the existence of a risk-neutral mutual fund (see, e.g., [Gornemann et al. \(2016\)](#)) that holds all interest-bearing assets—capital and public debt—and sells its shares to households.

Let total interest-bearing assets be denoted by $A_t = K_t + B_t$. The zero-profit condition of the mutual fund implies:

$$\tilde{r}_t A_{t-1} = \tilde{r}_t^k K_{t-1} + \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1\right) B_{t-1}. \quad (9)$$

Since capital and public debt are perfect substitutes, a no-arbitrage condition must hold: the expected real return on government bonds must equal the expected real return on capital. Formally,

$$\mathbb{E}_t \left[\frac{\tilde{R}_t^n}{\Pi_{t+1}} \right] = \mathbb{E}_t [1 + \tilde{r}_{t+1}^k]. \quad (10)$$

2.5 Government and its tools

We assume the existence of a benevolent government that chooses a sequence of public expenditures $(G_t)_{t \geq 0}$. The government finances this spending using several instruments. First, it can issue one-period nominal public debt B_t , as described earlier. We assume the existence of an enforcement technology that guarantees the debt is default-free. Second, the government raises revenues through distortionary taxes on labor income and capital returns.

The labor income tax, denoted by $\mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t})$ for pre-tax labor income $\tilde{w}_t y_{i,t} l_{i,t}$, is assumed to be possibly nonlinear and time-varying. Following [Heathcote et al. \(2017\)](#) (hence-

forth, HSV), it is defined as:

$$\mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t}) = \tilde{w}_t y_{i,t} l_{i,t} - \kappa_t (\tilde{w}_t y_{i,t} l_{i,t})^{1-\tau_t}, \quad (11)$$

where κ_t governs the overall level of taxation and τ_t captures the progressivity of the tax schedule. Both parameters are policy instruments and may vary over time. When $\tau_t = 0$, the labor income tax is linear with rate $1 - \kappa_t$; in contrast, $\tau_t = 1$ corresponds to full income redistribution, where all agents receive the same post-tax income κ_t .¹

Capital income is taxed linearly at rate τ_t^k . Total government revenues therefore equal:

$$\int_i \mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t}) \ell(di) + \tau_t^k \tilde{r}_t^k K_{t-1} + \tau_t^k \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right) B_{t-1}.$$

Given these instruments, the government budget constraint is:

$$G_t + \frac{\tilde{R}_{t-1}^n}{\Pi_t} B_{t-1} = D_t + \int_i \mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t}) \ell(di) + \tau_t^k \tilde{r}_t^k K_{t-1} + \tau_t^k \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right) B_{t-1} + B_t. \quad (12)$$

Substituting (11) into (12) and using the profit identity (8) together with $Y_t \xi_t = (\tilde{r}_t^k + \delta) K_{t-1} + \tilde{w}_t L_t$ yields the government budget constraint expressed in terms of pre-tax variables. Defining the post-tax quantities:

$$r_t = (1 - \tau_t^k) \tilde{r}_t, \quad r_t^k = (1 - \tau_t^k) \tilde{r}_t^k, \quad \frac{R_t^n}{\Pi_t} = (1 - \tau_t^k) \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right), \quad w_t = \kappa_t (\tilde{w}_t)^{1-\tau_t}, \quad (13)$$

and using the asset-income identity $r_t A_{t-1} = r_t^k K_{t-1} + \frac{R_t^n}{\Pi_t} B_{t-1}$, the budget constraint can be written compactly as:

$$G_t + B_{t-1} + r_t A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t - \delta K_{t-1} + B_t. \quad (14)$$

The intermediate algebra is provided in Appendix A.

2.6 Agents' program, resource constraints, and equilibrium

Agents' program. Consider an agent $i \in \mathcal{I}$ whose income consists of labor earnings and returns on savings. Post-tax labor income for an agent with productivity $y_{i,t}$ and labor effort $l_{i,t}$ is given by $\tilde{w}_t y_{i,t} l_{i,t} - \mathcal{T}_{i,t}(\tilde{w}_t y_{i,t} l_{i,t}) = w_t (y_{i,t} l_{i,t})^{1-\tau_t}$. Because a mutual fund aggregates all capital and public debt, agents face no portfolio choice. Let $a_{i,t}$ denote agent i 's holdings

¹The literature typically adopts either a linear tax combined with a lump-sum transfer (e.g., [Dyrda & Pedroni \(2022\)](#); [Açikgöz et al. \(2022\)](#)) or the non-linear tax structure proposed by [Heathcote et al. \(2017\)](#) (HSV). Recent quantitative work, such as [Heathcote & Tsujiyama \(2021\)](#), shows that the HSV specification provides a more realistic representation of labor income taxation.

of fund shares at time t . Borrowing is limited by $a_{i,t} \geq -\bar{a}$, where $\bar{a} > 0$ represents the borrowing limit. Fund claims yield the post-tax interest rate r_t , so savings payoffs are $(1 + r_t)a_{i,t-1}$, with $a_{i,t-1}$ denoting asset holdings at the end of period $t - 1$.

Each agent allocates resources between consumption, savings, and labor supply. Given the exogenous path of public expenditure $(G_t)_{t \geq 0}$, the optimization problem is:

$$\max_{\{c_{i,t}, l_{i,t}, a_{i,t}\}_{t=0}^{\infty}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t [u(c_{i,t}) - v(l_{i,t})], \quad (15)$$

$$\text{s.t.} \quad c_{i,t} + a_{i,t} \leq w_t(y_{i,t}l_{i,t})^{1-\tau_t} + (1 + r_t)a_{i,t-1}, \quad (16)$$

$$a_{i,t} \geq -\bar{a}, \quad c_{i,t} > 0, \quad l_{i,t} > 0. \quad (17)$$

Expectations $\mathbb{E}_0$ are taken over idiosyncratic and aggregate shocks, given the initial state $(a_{i,-1}, y_{i,0})$. At date 0, the agent chooses consumption $(c_{i,t})_{t \geq 0}$, labor $(l_{i,t})_{t \geq 0}$, and savings $(a_{i,t})_{t \geq 0}$ to maximize utility (15) subject to constraints (16)–(17). Decisions depend on $(a_{i,-1}, y_{i,0})$, the idiosyncratic history y_i^t , and the aggregate history z^t .

In period 0, each agent draws initial assets and productivity $(a_{i,-1}, y_{i,0})$ from the distribution $\mathbf{\Lambda}_0(a, y)$ defined on the Borel sets of $[-\bar{a}, \infty) \times \mathcal{Y}$, with $\mathbf{\Lambda}_0(a, y) : [-\bar{a}, \infty) \times \mathcal{Y} \rightarrow \mathbb{R}^+$. This setup allows the economy to start from any arbitrary, possibly stationary, distribution.

Remark 1 (Simplifying notation). *For an agent with initial state $(a_{i,-1}, y_{i,0})$ and idiosyncratic history y_i^t in aggregate state z^t , we denote the realization of any random variable $X_t((a_{i,-1}, y_{i,0}), y_i^t, z^t)$ simply by $X_{i,t}$.*

Aggregate variables are obtained by integrating over agents:

$$\int_i X_{i,t} \ell(di),$$

which is equivalent to the full sequential representation integrating over initial states (measure Λ) and idiosyncratic histories (measure θ):

$$\sum_{y^t \in \mathcal{Y}^{t+1}} \sum_{y_0 \in \mathcal{Y}} \int_{a_{-1} \in [-\bar{a}, \infty)} \theta_t(y^t) X_t((a_{-1}, y_0), y^t, z^t) \mathbf{\Lambda}(da_{-1}, y_0).$$

There thus exist functions $(c_t, l_t, a_t)_{t \geq 0}$ defined on $([-\bar{a}, \infty) \times \mathcal{Y}) \times \mathcal{Y}^{t+1} \times \mathbb{R}^{t+1}$ such that:

$$c_{i,t} = c_t((a_{i,-1}, y_{i,0}), y_i^t, z^t), \quad l_{i,t} = l_t((a_{i,-1}, y_{i,0}), y_i^t, z^t), \quad a_{i,t} = a_t((a_{i,-1}, y_{i,0}), y_i^t, z^t).$$

In what follows, we suppress functional arguments and keep the index i for brevity, following Remark 1.²

²Existence of these policy functions follows from standard results in [Açikgöz \(2016\)](#), [Cheridito & Sagredo \(2016\)](#), and [Miao \(2006\)](#).

The optimality conditions for problem (15)–(17) are:

$$u'(c_{i,t}) = \beta \mathbb{E}_t[(1 + r_{t+1})u'(c_{i,t+1})] + \nu_{i,t}, \quad (18)$$

$$v'(l_{i,t}) = (1 - \tau_t)w_t y_{i,t}(y_{i,t}l_{i,t})^{-\tau_t}u'(c_{i,t}), \quad (19)$$

where $\nu_{i,t}$ is the (discounted) Lagrange multiplier on the borrowing constraint. When the constraint does not bind, $\nu_{i,t} = 0$.

Market clearing and resource constraints. Market clearing requires:

$$\int_i a_{i,t} \ell(di) = A_t = B_t + K_t, \quad \int_i y_{i,t} l_{i,t} \ell(di) = L_t. \quad (20)$$

For simplicity, these conditions are written as integrals over agents i , though they are equivalent to integrating over initial states and idiosyncratic histories.³

The aggregate resource constraint is obtained by integrating the household budget constraint (16) over all agents,

$$C_t + A_t = w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + (1 + r_t)A_{t-1},$$

and combining this expression with the government budget constraint (14) and the identity $A_t = K_t + B_t$. This yields the economy-wide resource constraint:

$$C_t + G_t + K_t = \left(1 - \frac{\psi}{2}(\Pi_t - 1)^2\right) Y_t + K_{t-1} - \delta K_{t-1}, \quad (21)$$

where $C_t = \int_i c_{i,t} \ell(di)$ denotes aggregate consumption.

Equilibrium definition. The market equilibrium is defined as follows.

Definition 1 (Sequential equilibrium). *A sequential competitive equilibrium is a collection of individual allocations $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$; aggregate quantities $(K_t, L_t, Y_t, D_t, \xi_t)_{t \geq 0}$; price processes $(r_t, \tilde{r}_t, r_t^k, \tilde{r}_t^k, R_t^n, \tilde{R}_t^n, w_t, \tilde{w}_t)_{t \geq 0}$; fiscal policies $(\tau_t^k, \tau_t, \kappa_t, B_t, G_t)_{t \geq 0}$; and a monetary policy $(\Pi_t)_{t \geq 0}$, such that, for an initial distribution of wealth and productivity $(a_{i,-1}, y_{i,0})_{i \in \mathcal{I}}$, initial stocks of capital and public debt satisfying $K_{-1} + B_{-1} = \int_i a_{i,-1} \ell(di)$, and an initial aggregate shock z_0 , the following conditions hold:*

1. **Household optimization.** *Given prices, $(c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{t \geq 0, i \in \mathcal{I}}$ solve the agent's optimization problem (15)–(17).*

³Using the sequential representation, $\int_i a_{i,t} \ell(di) = \sum_{y^t \in \mathcal{Y}^{t+1}} \sum_{y_0 \in \mathcal{Y}} \int_{a_{-1} \in [-\bar{a}, \infty)} \theta_t(y^t) a_t((a_{-1}, y_0), y^t, z^t) \mathbf{\Lambda}(da_{-1}, y_0) = A_t(z^t) = A_t$, where dependence on z^t is omitted for brevity.

2. **Market clearing.** Financial, labor, and goods markets clear at all dates: for every $t \geq 0$, conditions (20) and (21) hold.
3. **Government budget constraint.** For every $t \geq 0$, the government budget constraint (14) is satisfied.
4. **Factor prices and asset returns.** Factor prices $(r_t, \tilde{r}_t, r_t^k, \tilde{r}_t^k, R_t^n, \tilde{R}_t^n, w_t, \tilde{w}_t)_{t \geq 0}$ are consistent with the production relationship (5), the asset-pricing conditions (9) and (10), and the post-tax definitions (13).
5. **Price adjustment and inflation.** The inflation path $(\Pi_t)_{t \geq 0}$ satisfies the New Keynesian Phillips curve (7) for all $t \geq 0$.

3 The Ramsey Problem

3.1 The Ramsey program.

First, define the social welfare function as the expected discounted sum of individual utilities across all agents:

$$W_0 = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} (u(c_{i,t}) - v(l_{i,t})) \ell(di) \right], \quad (22)$$

where $\omega_{i,t}$ represents the Pareto weight of agent i in period t . For a detailed discussion of the choice and interpretation of Pareto weights, see [Le Grand et al. \(2025\)](#).

In the Ramsey program, the planner determines a fiscal and monetary policy consistent with a competitive equilibrium that maximizes aggregate welfare according to the criterion in equation (22), while satisfying the government budget constraint. A **Ramsey equilibrium** is a set of fiscal policies, prices, individual allocations, and aggregate quantities that solve the Ramsey program. A **Ramsey steady-state equilibrium** is a time-invariant Ramsey

equilibrium. Formally, the Ramsey program can be stated as follows.

$$\max_{(r_t, \tilde{r}_t, r_t^k, \tilde{r}_t^k, R_t^n, \tilde{R}_t^n, w_t, \tilde{w}_t, \tau_t^k, \tau_t, \kappa_t, B_t, G_t, \Pi_t, K_t, L_t, Y_t, D_t, \xi_t, (c_{i,t}, l_{i,t}, a_{i,t}, \nu_{i,t})_{i \in \mathcal{I}})_{t \geq 0}} W_0, \quad (23)$$

$$G_t + B_{t-1} + r_t A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2\right) Y_t - \delta K_{t-1} + B_t, \quad (24)$$

$$\text{for all } i \in \mathcal{I}: \quad c_{i,t} + a_{i,t} = (1 + r_t) a_{i,t-1} + w_t (y_{i,t} l_{i,t})^{1-\tau_t}, \quad (25)$$

$$a_{i,t} \geq -\bar{a}, \quad \nu_{i,t} (a_{i,t} + \bar{a}) = 0, \quad \nu_{i,t} \geq 0, \quad (26)$$

$$u'(c_{i,t}) = \beta \mathbb{E}_t \left[(1 + r_{t+1}) u'(c_{i,t+1}) \right] + \nu_{i,t}, \quad (27)$$

$$v'(l_{i,t}) = (1 - \tau_t) w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t}), \quad (28)$$

$$\Pi_t (\Pi_t - 1) = \frac{\varepsilon - 1}{\psi} (\xi_t - 1) + \beta \mathbb{E}_t \left[\Pi_{t+1} (\Pi_{t+1} - 1) \frac{Y_{t+1}}{Y_t} \frac{M_{t+1}}{M_t} \right], \quad (29)$$

$$\int_i a_{i,t} \ell(di) = A_t = K_t + B_t, \quad L_t = \int_i y_{i,t} l_{i,t} \ell(di), \quad (30)$$

$$\tilde{r}_t A_{t-1} = \tilde{r}_t^k K_{t-1} + \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right) B_{t-1}, \quad (31)$$

$$\mathbb{E}_t \left[\frac{\tilde{R}_t^n}{\Pi_{t+1}} \right] = \mathbb{E}_t [1 + \tilde{r}_{t+1}^k], \quad (32)$$

$$w_t = \kappa_t (\tilde{w}_t)^{1-\tau_t}, \quad (33)$$

$$r_t^k = (1 - \tau_t^k) \tilde{r}_t^k, \quad (34)$$

$$\xi_t = \frac{\tilde{r}_t^k + \delta K_{t-1}}{\alpha Y_t}, \quad (35)$$

$$M_t = \int_i u'(c_{i,t}) \ell(di), \quad (36)$$

$$Y_t = \left(\int_i a_{i,t-1} \ell(di) - B_{t-1} \right)^\alpha \left(\int_i y_{i,t} l_{i,t} \ell(di) \right)^{1-\alpha}. \quad (37)$$

Since the planner chooses policies and allocations to maximize welfare subject to the constraints of the economy, the individual budget constraints (25)–(28) are explicitly included in the Ramsey program. The government budget constraint (24), the New Keynesian Phillips curve (29), the aggregation conditions (30), the non-profit condition (31), the non-arbitrage condition (32), the post-tax definitions (33)–(34), the definition of ξ_t (35), the definition of M_t (36), and the production function (37) are also explicitly imposed.⁴ The planner

⁴The term $M_t = \int_i u'(c_{i,t}) \ell(di)$ represents the *aggregate marginal utility of consumption*, which serves as the welfare-based stochastic discount factor in the planner's problem. Defining M_t as the integral of individual marginal utilities is standard in heterogeneous-agent Ramsey frameworks (see, e.g., [Açikgöz et al. \(2022\)](#); [Heathcote & Tsujiyama \(2021\)](#)). It ensures that aggregate intertemporal trade-offs are weighted by the social valuation of individual consumption units, thereby linking welfare aggregation and the pricing kernel in the planner's problem. It aggregates individual consumption marginal utilities into a single measure that summarizes the intertemporal value of resources from the planner's perspective. In equilibrium, M_t

chooses sequences of prices, fiscal policies, individual allocations, and aggregate quantities to maximize welfare (23) subject to these constraints. The instruments available to the planner are: the labor income tax parameters (τ_t, κ_t) , the capital income tax rate τ_t^k , the nominal public debt B_t , the government expenditure G_t , and the gross inflation rate Π_t .

Let μ_t denote the Lagrange multiplier on the government budget constraint (24), γ_t the multiplier on the Phillips curve (29), Γ_t the multiplier on the non-profit condition (31), and Ψ_t the multiplier on the non-arbitrage condition (32).

3.2 Interpretation of the Ramsey optimality conditions

The economic trade-offs faced by the planner can be understood from the first-order conditions of the Ramsey program, reported in Appendix B.⁵ These conditions jointly characterize how optimal fiscal and monetary policy reconcile nominal rigidities, intertemporal smoothing, and distributional efficiency in an economy with heterogeneous agents. In what follows, we interpret the main optimality conditions, highlighting how the associated multipliers and wedges shape the planner's intertemporal and cross-sectional trade-offs in the optimal allocation.

The inflation condition captures the central trade-off of monetary policy in the presence of price adjustment costs. It can be written as:

$$\underbrace{\mu_t \psi (\Pi_t - 1)}_{\text{total cost due to price dispersion}} = \underbrace{(\gamma_{t-1} - \gamma_t)(2\Pi_t - 1)M_t}_{\text{real-wage manipulation}} + \underbrace{\left(\Gamma_t(1 - \tau_t^k)B_{t-1} - \beta^{-1}\Psi_{t-1} \right) \frac{\tilde{R}_{t-1}^n}{\Pi_t^2 Y_t}}_{\text{reduction of real interest payments on debt}}, \quad (38)$$

which equates the marginal cost of inflation to the sum of two benefits: the ability to reallocate resources between wage setters and households through nominal rigidities, and the fiscal advantage of eroding the real value of nominal debt. The multiplier μ_t reflects the shadow value of relaxing the government's budget constraint, while γ_t captures the value of distorting prices through the Phillips curve. Inflation thus serves as a policy instrument balancing fiscal consolidation and wage adjustment incentives. In the limiting case of flexible prices ($\psi \rightarrow 0$) and no nominal debt ($B_t = 0$), inflation ceases to have any real allocative role, and this condition becomes vacuous.

The link between nominal and real returns is governed by two complementary conditions

enters the New Keynesian Phillips curve (29) and the Ramsey first-order conditions as the object that translates welfare costs into nominal distortions. Intuitively, heterogeneity in consumption and marginal utilities across agents affects M_t , amplifying the welfare cost of price dispersion and influencing the planner's valuation of current versus future resources.

⁵Solving the Ramsey problem through a Lagrangian formulation involves several technical subtleties, discussed in detail in [LeGrand & Ragot \(2025\)](#).

ensuring internal policy consistency:

$$0 = -\beta\Gamma_{t+1}(1 - \tau_{t+1}^k)\frac{B_t}{\Pi_{t+1}} + \Psi_t\frac{1}{\Pi_{t+1}}, \quad (39)$$

and:

$$\beta^{-1}\Psi_{t-1} = \left[\left(\frac{\epsilon - 1}{\psi} \right) \frac{\gamma_t}{\alpha} M_t - \Gamma_t(1 - \tau_t^k) \right] K_{t-1}. \quad (40)$$

The first condition links the shadow values of bond and capital returns through Ψ_t , ensuring that expected real yields are consistent across nominal assets. The second expresses how the planner's valuation of capital depends on price rigidity and the markup gap, $(\epsilon - 1)/\psi$, with Γ_t ensuring that the pre-tax capital return is consistent with the aggregate return on wealth. When there are no wedges—no capital taxation, stable prices, and efficient markups—these two equations jointly reproduce the standard no-arbitrage condition equating real returns across assets.

The intertemporal allocation problem for individual agents is characterized:

$$\begin{aligned} \psi_{i,t} = & \beta\mathbb{E}_t[(1 + r_{t+1})\psi_{i,t+1}] + \beta\mathbb{E}_t\left[\mu_{t+1}\left(\left(1 - \frac{\psi}{2}(\Pi_{t+1} - 1)^2\right)\frac{\tilde{r}_{t+1}^k + \delta}{\xi_{t+1}} - r_{t+1} - \delta\right)\right] \quad (41) \\ & + \left(\frac{\epsilon - 1}{\psi}\right)\beta\mathbb{E}_t\left[\gamma_{t+1}(\xi_{t+1} - \alpha)M_{t+1}\frac{Y_{t+1}}{K_t}\right] - \beta\mathbb{E}_t\left[(\gamma_{t+1} - \gamma_t)\Pi_{t+1}(\Pi_{t+1} - 1)\alpha\frac{M_{t+1}Y_{t+1}}{K_t}\right] \\ & - \beta\mathbb{E}_t\left[\Gamma_{t+1}(1 - \tau_{t+1}^k)\tilde{r}_{t+1}^k\right] + \beta\mathbb{E}_t\left[\Gamma_{t+1}r_{t+1}\right], \end{aligned}$$

which governs the evolution of the shadow value $\psi_{i,t}$ associated with relaxing the individual budget constraint. This condition extends the standard intertemporal optimality relation of the representative-agent Ramsey problem to a setting with heterogeneous agents and nominal frictions. The first term, $\beta\mathbb{E}_t[(1 + r_{t+1})\psi_{i,t+1}]$, reflects the continuation value of wealth accumulation through private savings. The second term, weighted by μ_{t+1} , adjusts this value by the planner's shadow price of public funds and by the effective return on aggregate resources in the presence of nominal rigidities. The next two terms capture the welfare effects of price-setting distortions: the first arises from markup deviations from the efficient level $(\xi_{t+1} - \alpha)$, while the second accounts for inflation dynamics and price dispersion through the multipliers $(\gamma_t - \gamma_{t+1})$. Finally, the Γ_t terms measure the discrepancy between the private and social valuations of returns on capital and bonds, thus connecting the individual saving decision to aggregate asset-pricing consistency.

Rewriting this condition as:

$$\hat{\psi}_{i,t} = \beta\mathbb{E}_t\left[\hat{\psi}_{i,t}(1 + r_{t+1})\right] + \beta\mathbb{E}_t\left[\Gamma_{t+1}\left(r_{t+1} - (1 - \tau_{t+1}^k)\left(\frac{\tilde{R}_t^n}{\Pi_{t+1}} - 1\right)\right)\right], \quad (42)$$

where $\hat{\psi}_{i,t} = \psi_{i,t} - \mu_t$, highlights how heterogeneity in the marginal value of relaxing the individual constraint, $\hat{\psi}_{i,t}$, shapes the planner's intertemporal allocation of resources. Agents

with higher $\hat{\psi}_{i,t}$ —that is, those facing tighter budget constraints or higher social weights—are prioritized in the planner’s optimal redistribution across periods and states. The second term, involving Γ_{t+1} , shows how deviations between the after-tax return on capital and the inflation-adjusted bond return introduce allocative wedges. These wedges reflect the joint influence of fiscal and monetary policy on the intertemporal distribution of resources and ultimately determine how efficiency and equity considerations are balanced in the Ramsey allocation.

The first-order condition with respect to public debt,

$$\begin{aligned} \mu_t = & \beta \mathbb{E}_t \left[\mu_{t+1} \left(1 - \delta + \left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \frac{\tilde{r}_{t+1}^k + \delta}{\xi_{t+1}} \right) \right] \\ & + \beta \mathbb{E}_t \left[\left(\left(\frac{\epsilon - 1}{\psi} \right) \gamma_{t+1} (\xi_{t+1} - \alpha) - \alpha (\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \right) \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ & + \beta \mathbb{E}_t \left[\Gamma_{t+1} (1 - \tau_{t+1}^k) \left(\frac{\tilde{R}_t^n}{\Pi_{t+1}} - 1 - \tilde{r}_{t+1}^k \right) \right], \end{aligned} \quad (43)$$

extends the standard Euler equation for μ_t by incorporating inflation distortions, markup gaps, and tax wedges. This condition governs the optimal intertemporal smoothing of fiscal resources: when frictions vanish, it simplifies to $\mu_t = \beta(1 + r_{t+1})\mathbb{E}_t[\mu_{t+1}]$, the canonical result that debt equates the marginal cost of public funds across periods.

Labor market distortions emerge from the wedge between the marginal rate of substitution and the marginal product of labor, expressed as:

$$\begin{aligned} \frac{\omega_{i,t} v'(l_{i,t}) + \lambda_{i,l,t} v''(l_{i,t})}{(1 - \tau_t) w_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t}} = & \hat{\psi}_{i,t} - \lambda_{i,l,t} \tau_t \frac{u'(c_{i,t})}{l_{i,t}} + \mu_t \frac{(1 - \frac{\psi}{2} (\Pi_t - 1)^2) F_{L,t}}{(1 - \tau_t) w_t (y_{i,t})^{-\tau_t} (l_{i,t})^{-\tau_t}} \\ & + \left[\left(\frac{\epsilon - 1}{\psi} \right) \gamma_t (\xi_t - (1 - \alpha)) - (1 - \alpha) (\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) \right] \frac{Y_t M_t}{L_t (1 - \tau_t) w_t (y_{i,t})^{-\tau_t} (l_{i,t})^{-\tau_t}}, \end{aligned} \quad (44)$$

where the first term captures the heterogeneity in agents’ effective labor distortions ($\hat{\psi}_{i,t}$), the second term reflects progressivity effects of labor taxation, and the last two terms capture the influence of inflation dynamics and markup fluctuations on labor allocation. The planner adjusts w_t and τ_t to manage these wedges at both the individual and aggregate level, as summarized by:

$$0 = \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{j,l,t} (1 - \tau_t) u'(c_{j,t}) / l_{j,t} \right) \ell(dj) + \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t M_t \frac{1}{1 - \tau_t} \left(\frac{w_t}{\kappa} \right)^{\frac{\tau_t}{1-\tau_t}} \frac{1}{\kappa} \frac{L_t}{1 - \alpha}, \quad (45)$$

and:

$$\begin{aligned}
0 = & \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{j,l,t} (1 - \tau_t) (u'(c_{j,t})/l_{j,t}) \right) \ln(y_{j,t} l_{j,t}) \ell(dj) \\
& + \int_j \lambda_{j,l,t} (u'(c_{j,t})/l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) - \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t M_t \frac{1}{(1 - \tau_t)^2} \left(\frac{w_t^{\frac{\tau_t}{1-\tau_t}}}{\kappa^{\frac{1}{1-\tau_t}}} \right) \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{(1 - \alpha)},
\end{aligned} \tag{46}$$

which describe, respectively, how aggregate wage-setting and tax progressivity affect the propagation of heterogeneity in the economy.

Finally, the equilibrium return on assets satisfies:

$$0 = \int_j \left(\hat{\psi}_{j,t} a_{j,t-1} + \lambda_{j,c,t-1} u'(c_{j,t}) \right) \ell(dj) + \Gamma_t (B_{t-1} + K_{t-1}), \tag{47}$$

ensuring consistency between the after-tax return and the distribution of asset holdings and marginal utilities. This condition guarantees that the return on assets aligns with the planner's valuation of resources in the presence of capital taxation and nominal distortions.

Taken together, these first-order conditions summarize how the planner optimally balances price adjustment costs, fiscal distortions, and distributional objectives over time. For a comprehensive theoretical discussion of the structure and interpretation of these conditions, see [LeGrand & Ragot \(2025\)](#).

3.3 Identification of the Social Welfare Function

Our identification strategy is to recover the Pareto weights $(\omega_{i,t})$ that are consistent with the first-order conditions of the Ramsey planner—namely, equations (53) to (64) in Appendix B—and that are closest to the utilitarian benchmark. This approach allows us to identify the welfare weights that rationalize the observed fiscal policies and the resulting distributional outcomes within a given economy.

We proceed under the assumption that the observed fiscal and monetary policies correspond to the optimal choices of a benevolent social planner who maximizes intertemporal welfare in a heterogeneous-agent economy according to the welfare function:

$$W_0 = \mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} (u(c_{i,t}) - v(l_{i,t})) \ell(di) \right]. \tag{48}$$

The planner is endowed with a Social Welfare Function (SWF) and internalizes all distortions and general-equilibrium effects associated with fiscal and monetary instruments. Our empirical identification therefore consists in estimating the sequence of Pareto weights $\omega_{i,t}$ that make the system of first-order conditions hold for the observed allocations and policies.

Since multiple welfare functions can, in principle, rationalize the same set of fiscal outcomes, we introduce an additional identification criterion: among all admissible SWFs consistent with the Ramsey optimality conditions, we select the one that is *closest* to the Utilitarian benchmark. Formally, this corresponds to minimizing a quadratic distance between the estimated weights and the uniform distribution, that is,

$$\min_{\{\omega_{i,t}\}} \sum_{i,t} (\omega_{i,t} - 1)^2, \quad (49)$$

subject to the planner’s optimality conditions. This ensures that the inferred welfare function departs from the Utilitarian benchmark only insofar as required to rationalize the observed policies and inequality patterns. For a detailed discussion of this identification approach and its normative interpretation, see [Le Grand et al. \(2025\)](#).

4 Quantitative Investigation

In this paper, our objectives are twofold. First, we construct calibrated models that replicate the economic environment of the United Kingdom in two benchmark years: 2007 and 2021. The year 2007 serves as a pre-crisis reference point, preceding both the global financial crisis and the COVID-19 pandemic—two events that profoundly reshaped fiscal structures. In contrast, 2021 captures the post-pandemic landscape, marked by persistently high public debt and structural changes in fiscal policy. We interpret these sustained debt levels as evidence of a possible transition toward a new steady state.

Second, rather than focusing on the optimality of the tax system per se, we study the dynamics of real variables, public debt, and inflation in response to shocks. We begin by calibrating the model parameters to reproduce realistic steady states consistent with the fiscal policies and observed inequalities of the UK in 2007 and 2021. Building on the literature on the inverse taxation problem, we then estimate the underlying Social Welfare Function (SWF) that rationalizes the observed fiscal configuration as the planner’s optimal policy in the steady state.

This calibration strategy allows us to examine how fiscal instruments and macroeconomic variables adjust dynamically following transitory shocks, starting from realistic period-0 allocations. By introducing disturbances to the economy, we analyze how fiscal tools, real activity, public debt, inflation, and capital evolve over time, verifying that policy variables converge back to the assumed optimal steady state. Finally, we explore how inflation responds when fiscal policy instruments are absent, highlighting potential asymmetries between the UK’s 2021 and 2007 economic environments.

The next section details the calibration procedure for the UK economy in both benchmark years.

4.1 The UK economy in 2007

The estimated parameters are summarized in Table 2, and the calibration strategy is detailed below.

Preference parameters. The model is calibrated at a quarterly frequency. The discount factor is set to $\beta = 0.99$ to match a realistic capital-to-output ratio (K/Y). The period utility function in equation (1) is specified as $u(c_{i,t}) = \frac{c_{i,t}^{1-\gamma}-1}{1-\gamma}$ with $\gamma = 2$, and $v(l_{i,t}) = \frac{1}{\chi} \frac{l_{i,t}^{1+\frac{1}{\varphi}}}{1+\frac{1}{\varphi}}$. The Frisch elasticity of labor supply is set to $\varphi = 0.5$, following Chetty et al. (2011) for the intensive margin. The labor-scaling parameter is $\chi = 0.058$, implying an aggregate labor supply normalized to 0.39.

Technology and TFP shocks. Production follows a Cobb–Douglas specification, $Y_t = Z_t K_{t-1}^\alpha L_t^{1-\alpha}$, with a capital share $\alpha = 0.36$ and a depreciation rate $\delta = 2.5\%$. Total factor productivity evolves according to a standard AR(1) process:

$$Z_t = Z_0 e^{z_t}, \quad z_t = \rho_z z_{t-1} + \varepsilon_{z,t}, \quad \varepsilon_{z,t} \sim_{\text{iid}} \mathcal{N}(0, \sigma_z^2).$$

In the quantitative exercises below, we vary the parameters of the TFP process to explore the effects of productivity shocks.

Idiosyncratic labor risk and taxation. Idiosyncratic productivity follows an AR(1) process,

$$\log y_t = \rho_y \log y_{t-1} + \varepsilon_{y,t}, \quad \varepsilon_{y,t} \sim_{\text{iid}} \mathcal{N}(0, \sigma_y^2),$$

where (ρ_y, σ_y) are estimated to match key empirical moments of the UK economy in 2007. We also calibrate the parameters (τ, κ) governing the progressivity of labor income taxation. The target moments are (i) the debt-to-GDP ratio (B/Y), (ii) the post-tax Gini coefficient, and (iii) the variance of pre-tax income.

Using data from the Office for National Statistics (ONS), we set the target debt-to-GDP ratio to 43.1% and the post-tax Gini coefficient to 0.386. The variance of pre-tax income, corresponding to a Gini coefficient of 0.535, is also drawn from ONS data. The parameters are estimated via the Simulated Method of Moments to match these statistics. The resulting calibration yields $\rho_y = 0.995$ and $\sigma_y = 0.07$. The estimated tax progressivity parameters are $\tau_t = 0.05$ and $\kappa_t = 0.62$. The capital income tax rate is set at $\tau_t^k = 38\%$, based on Bank of England estimates. The AR(1) process is discretized using the procedure of Rouwenhorst (1995) with seven states.

The model reproduces the targeted statistics closely: it generates a pre-tax Gini of 0.53 (target 0.535), a post-tax Gini of 0.384 (target 0.386), and a debt-to-GDP ratio of 42.5%

(target 43.1%), as reported in Table 1. It also implies a government spending-to-GDP ratio of 27.6% and total tax revenues of 29.3% of GDP.

In addition, the model predicts a consumption-to-GDP ratio of 46.8% and a consumption variance of 29.2%. The implied wealth Gini is 0.689. These results, summarized in Table 3, indicate that the calibrated tax system provides a close approximation to the redistributive effects of the actual UK fiscal system and successfully replicates key macroeconomic and distributional statistics, consistent with the findings of Heathcote et al. (2017) and Dyrda & Pedroni (2022).

4.2 The UK economy in 2021

The estimated economy of the United Kingdom in 2021 shares many structural features with that of 2007. For consistency, we maintain the same quarterly frequency and functional forms.

Preference parameters. The model is quarterly. The discount factor is set to $\beta = 0.99$, and the Frisch elasticity of labor supply to $\varphi = 0.5$. Fixing the labor-scaling parameter to $\chi = 0.053$ normalizes aggregate labor supply to 0.35.

Technology and TFP shocks. Production follows a Cobb–Douglas specification,

$$Y_t = Z_t K_{t-1}^\alpha L_t^{1-\alpha},$$

with a capital share $\alpha = 0.36$ and depreciation rate $\delta = 2.5\%$. Total factor productivity evolves as:

$$Z_t = Z_0 e^{z_t}, \quad z_t = \rho_z z_{t-1} + \varepsilon_{z,t}, \quad \varepsilon_{z,t} \sim_{\text{iid}} \mathcal{N}(0, \sigma_z^2).$$

Idiosyncratic labor risk and taxation. Individual productivity follows an AR(1) process,

$$\log y_t = \rho_y \log y_{t-1} + \varepsilon_{y,t}, \quad \varepsilon_{y,t} \sim_{\text{iid}} \mathcal{N}(0, \sigma_y^2),$$

where (ρ_y, σ_y) are chosen to match key moments of the UK economy in 2021. As in the 2007 calibration, we jointly estimate the labor income tax parameters (τ, κ) to capture the observed progressivity of the tax system.

The calibration targets three moments: (i) the debt-to-GDP ratio (B/Y) , equal to 101% in 2021 according to Table 1; (ii) the post-tax and transfers Gini coefficient, 0.299; and (iii) the variance of pre-tax income, corresponding to a Gini of 0.502. All data are drawn from the Office for National Statistics (ONS).

Using the Simulated Method of Moments, we estimate parameters to match these targets. The resulting calibration yields $\rho_y = 0.992$ and $\sigma_y = 0.09$. The estimated tax parameters are $\tau = 0.22$ and $\kappa = 0.775$. The capital income tax rate is set to $\tau^k = 35\%$, consistent with Bank of England estimates. The AR(1) process is discretized using the method of [Rouwenhorst \(1995\)](#) with seven states.

The model reproduces the main empirical moments closely: it generates a pre-tax Gini of 0.50 (target 0.502), a post-tax Gini of 0.357 (target 0.299), and a debt-to-GDP ratio of 112% (target 101%). It also implies a government spending-to-GDP ratio of 14.6% and tax revenues equal to 19.2% of GDP. In addition, the model predicts a consumption-to-GDP ratio of 59.8%, a consumption variance of 19%, and a wealth Gini of 0.633. These quantitative implications, summarized in Tables 2 and 3, indicate that the model provides a realistic representation of the UK economy in 2021, capturing the main fiscal and distributional patterns of the post-pandemic environment.

Parameter	Description	UK 2007		UK 2021	
		Value	Target / Source	Value	Target / Source
<i>Preference parameters</i>					
β	Discount factor	0.99	$K/Y = 2.56$	0.99	$K/Y = 2.56$
γ	Risk aversion	2.0	Standard	2.0	Standard
φ	Frisch elasticity	0.5	Chetty et al. (2011)	0.5	Chetty et al. (2011)
χ	Labor scaling	0.058	Normalized $L = 0.39$	0.053	Normalized $L = 0.35$
<i>Technology and fiscal parameters</i>					
α	Capital share	0.36	ONS (Profit share)	0.36	ONS (Profit share)
δ	Depreciation rate	0.025	Standard	0.025	Standard
τ^k	Capital income tax	0.38	Bank of England	0.35	Bank of England
τ	Labor tax progressivity	0.05	Table 1	0.22	Table 1
κ	Labor tax scaling	0.62	Table 1	0.775	Table 1
<i>Idiosyncratic productivity parameters</i>					
ρ_y	Autocorrelation	0.995	Estimated	0.992	Estimated
σ_y	Std. deviation	0.07	Estimated	0.09	Estimated

Table 2: Calibration parameters for the UK economies in 2007 and 2021. Values differ only where fiscal or inequality conditions changed.

	UK 2007		UK 2021	
	Model	Data	Model	Data
B/Y	42.5%	43%	112%	101%
Gini (pre-tax income)	0.53	0.535	0.50	0.502
Gini (post-tax income)	0.384	0.386	0.357	0.299

Table 3: Model-implied moments for key fiscal and inequality indicators. Empirical values are discussed in Sections 4.1 and 4.2, and summarized in Table 1.

4.3 Numerical Tools and Estimation of Pareto Weights

As discussed above, our estimation procedure identifies Pareto weights that satisfy the first-order conditions of the social planner (i.e., equations (53)–(64) in Appendix B) and are as close as possible to the utilitarian benchmark. The methodology follows [Le Grand et al. \(2025\)](#). Below we provide a concise overview of the numerical tools used to solve the Ramsey problem defined by equations (23)–(37).

The main difficulty in identifying these weights arises because the Ramsey problem in Section 3.1 features a joint distribution over wealth and Lagrange multipliers. This joint object is high-dimensional and interacts with the planner’s instruments in a non-trivial way, making conventional perturbation methods around a steady state infeasible (see, e.g., [Reiter \(2009\)](#); [Boppart et al. \(2018\)](#); [Bayer et al. \(2019\)](#); [Auclert et al. \(2021\)](#)).

To overcome this challenge, we rely on the Lagrangian aggregation method of [Le Grand & Ragot \(2022a\)](#), further developed in subsequent work (see [Le Grand & Ragot \(2022b\)](#), [LeGrand & Ragot \(2023, 2025\)](#); [Le Grand et al. \(2025\)](#)). This approach computes the steady-state allocation and derives a finite set of equations that approximate, by perturbation, the dynamics of the Ramsey program under small aggregate shocks. The central idea is to aggregate the Bewley model—under a fixed policy and without aggregate uncertainty—by grouping together agents who share the same idiosyncratic history over the last N periods. Each group, or “aggregate” agent, is characterized by the average wealth and allocation of all individuals with this N -period history.

Formally, consider an agent i who at time t has the history $\{y_{i,0}, \dots, y_{i,t}\}$, and let $N \geq 0$ denote the truncation length. The aggregation step assigns identical allocations and wealth to all agents with the same idiosyncratic sequence over the last N periods. Such an N -period sequence is called a *truncated history*. For a full history $y^t = \{y_0, \dots, y_{t-N}, y_{t-N+1}, \dots, y_t\}$, the truncated representation is:

$$y^N = \{y_{t-N+1}^N, \dots, y_{t-1}^N, y_t^N\}.$$

We can write the history of agent i as:

$$y^t = \underbrace{\{y_0, \dots, y_{t-N}\}}_{\xi_{y^N}}, \underbrace{\{y_{t-N+1}^N, \dots, y_{t-1}^N, y_t^N\}}_{=y^N},$$

where the parameter ξ_{y^N} captures the residual heterogeneity associated with truncated history y^N . This parameter is constructed so that the truncated model exactly reproduces the aggregation of the underlying Bewley model in the absence of aggregate shocks. In practice, all agents with the same truncated history y^N are grouped into a single representative “aggregate” agent, and the model is expressed in terms of these finite groups.

This procedure yields the *truncated model*, a finite-dimensional representation of the economy. Within each truncated history, agents are assumed to share risks perfectly, effectively “forgetting” the heterogeneity that existed before the aggregation—see [LeGrand & Ragot \(2025\)](#). The parameter ξ_{y^N} allows this truncated model to map back to the original Bewley structure.⁶

Appendix C presents the truncated version of the nominal model introduced in Section 3.1, together with the planner’s first-order conditions for this case. At the steady state, these conditions can be expressed in closed form, allowing us to recover the Pareto weights using standard matrix algebra.⁷ The resulting weights are verified to satisfy the planner’s FOCs.

Following [Le Grand et al. \(2025\)](#), we adopt a truncation length of $N = 5$. Larger values of N leave the results virtually unchanged.⁸ For instance, with ten idiosyncratic productivity levels, the number of truncated histories equals $10^5 = 100,000$. Pareto weights are estimated so that histories with the same productivity level at the beginning of the truncation share the same weight, yielding ten distinct weights—one for each productivity level.

While straightforward to implement, this method includes many histories that are highly improbable and represent a negligible share of agents. To address this inefficiency, we employ the refined truncation approach of [Le Grand & Ragot \(2022b\)](#). This method replaces highly likely histories with extended ones that feature a larger truncation length. For example, the history (y_1, y_1) with $N = 2$ can be refined into $\{(y, y_1, y_1) : y \in \mathcal{Y}\}$, thereby splitting agents who have been at productivity y_1 for two consecutive periods into $\text{Card}(\mathcal{Y})$ distinct truncated histories. This refinement ensures that the total number of histories grows linearly, rather than exponentially, with the maximum truncation length—greatly improving computational tractability while preserving precision.

⁶[Le Grand & Ragot \(2018\)](#) and [Le Grand et al. \(2025\)](#) provide a detailed exposition of this methodology.

⁷For further details, see [Le Grand et al. \(2025\)](#), who apply a similar method to a model without monetary policy, using a parametric specification of the Social Welfare Function as in [Heathcote & Tsujiyama \(2021\)](#).

⁸[Le Grand & Ragot \(2022a\)](#) show that the truncated allocation converges to the true allocation as N increases, while [LeGrand & Ragot \(2023\)](#) demonstrate that relatively small truncation lengths deliver quantitatively accurate results.

4.4 Model Dynamics with Optimal Fiscal and Monetary Policy

Setup and method. We now turn to the dynamic implications of our calibrated model by simulating the economy’s response to aggregate shocks under alternative policy environments. To this end, we employ the numerical strategy outlined in Section 4.3 and developed in detail in Appendix C, which combines the truncation and perturbation methods to solve the Ramsey problem. Using this framework, we compute the dynamic equilibrium paths by simulating the economy’s response to aggregate shocks around the deterministic steady state of the truncated model. Specifically, we solve the Ramsey system (23)–(37) by applying a first-order perturbation to the steady state, expressing all variables in proportional deviations. The policy instruments include fiscal variables $(\tau_t^k, \tau_t, \kappa_t, B_t)$ and, when monetary policy is active, the additional instruments $(\Pi_t, \tilde{R}_t^n)$.

Sequence of experiments. We first analyze the case in which the planner has access only to fiscal tools, holding monetary policy passive. We then introduce an active monetary authority and assess how its interaction with fiscal policy alters the optimal adjustment paths. The analysis highlights how the initial debt level and shock persistence shape the planner’s optimal response.

4.4.1 Government spending shocks with fiscal policy only

We begin with a positive government spending shock, modeled as an autoregressive disturbance to public purchases:

$$g_t = \rho_g g_{t-1} + \varepsilon_{g,t},$$

with persistence parameter ρ_g and innovation $\varepsilon_{g,t}$. Figure 2 plots the responses of key macroeconomic variables for the UK economies of 2007 and 2021 under two persistence regimes: high persistence ($\rho_g = 0.97$) and low persistence ($\rho_g = 0.01$). The responses are expressed in percentage deviations from steady state.

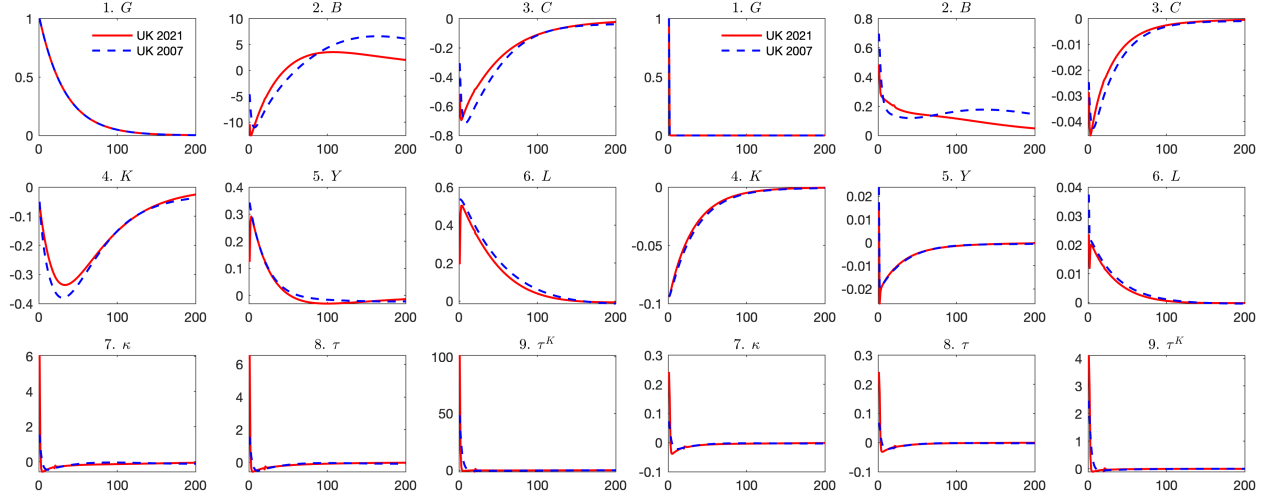


Figure 2: Impulse responses to a positive government spending shock under fiscal policy only. UK 2021 (red solid) and UK 2007 (blue dashed). Panel (a): high persistence ($\rho_g = 0.97$); Panel (b): low persistence ($\rho_g = 0.01$).

In both periods, the rise in government spending crowds out private investment, causing a temporary decline in capital K_t . This occurs because higher public expenditure reduces the need for private saving, leading to lower investment and a subsequent fall in the capital stock. Private consumption C_t also falls on impact as the planner reallocates resources toward public expenditure while attempting to smooth consumption intertemporally. However, the degree of persistence strongly influences the planner's optimal debt policy.

In the *low-persistence* scenario, the shock is short-lived, and the planner increases public debt B_t to provide households with a risk-free store of value. Debt issuance helps sustain consumption during the temporary fall in capital, thereby smoothing welfare costs over time. In contrast, under *high persistence*, the decline in capital is long-lasting, so additional debt issuance would require higher taxes in periods when the tax base is already weak (i.e., periods when the capital is still low). Anticipating this constraint, the planner restrains borrowing and opts for more immediate fiscal adjustment through higher taxes. The response of debt is therefore smaller and shorter-lived.

Comparing across calibrations, we observe that tax adjustments are substantially larger in 2021 than in 2007, reflecting the higher initial debt-to-GDP ratio and the smaller fiscal space available to accommodate shocks. Fiscal consolidation occurs earlier and more aggressively in 2021, leading to a smaller but more front-loaded debt response. This difference underscores how initial fiscal conditions shape the planner's optimal response to shocks, with high debt levels necessitating more immediate adjustments to maintain sustainability.

4.4.2 Government spending shocks with fiscal and monetary policy

We next allow the planner to simultaneously adjust both fiscal and monetary instruments. Figure 3 reports the impulse responses when the planner can manipulate inflation Π_t and the nominal interest rate $\tilde{R}_t^n$ in addition to fiscal variables.

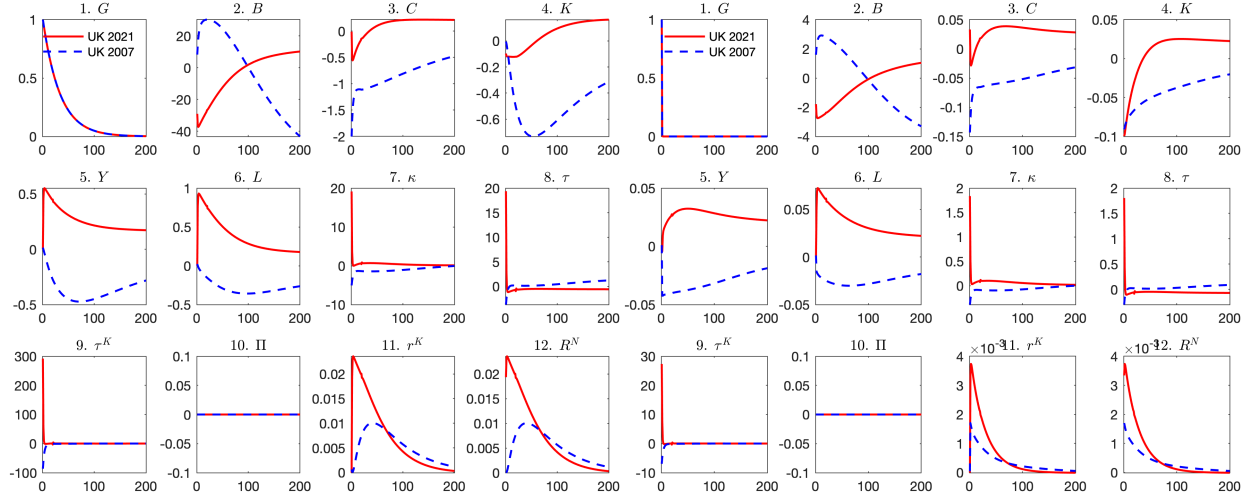


Figure 3: Positive government spending shock with optimal fiscal and monetary policy. UK 2021 (red solid) and UK 2007 (blue dashed). Panel (a): high persistence; Panel (b): low persistence.

Upon examining Figure 3, we consider a scenario in which the planner can employ both fiscal and monetary instruments to stabilize the economy following a positive government spending shock. When monetary policy is active, the overall adjustment dynamics change markedly relative to the case with fiscal instruments alone. A first striking feature is that the qualitative responses of macroeconomic aggregates are broadly comparable across high and low persistence shocks, reflecting the additional degree of freedom provided by monetary policy. Yet, the quantitative magnitudes and the composition of the adjustment differ substantially between the 2007 and 2021 calibrations.

In the case of the UK economy in 2021, the positive fiscal shock leads to a tightening of monetary conditions. To contain inflationary pressures, the planner raises nominal interest rates, which, in turn, depresses private investment and lowers the capital stock. The contraction in capital reduces the scope for intertemporal consumption smoothing, and the planner would ideally like to increase public debt to provide households with a safe asset and support aggregate demand. However, the elevated initial debt-to-GDP ratio limits this possibility: additional debt issuance would imply higher future taxes precisely when the economy operates with a reduced capital base. Anticipating this trade-off, the planner opts instead for an immediate increase in tax rates to stabilize debt, maintaining the intertemporal government budget constraint without amplifying long-term fiscal risks. The outcome is a front-loaded

fiscal adjustment that offsets the rise in interest payments and preserves debt sustainability, at the cost of a sharper short-run contraction in private demand.

In contrast, the 2007 calibration displays a markedly more countercyclical fiscal stance. Despite the decline in private capital induced by higher government expenditure, public debt increases on impact as the government borrows to finance the temporary spending expansion. This rise in debt reflects two reinforcing mechanisms. First, the larger fall in private capital observed in 2007 weakens investment and output, generating a temporary recession. Second, the fiscal expansion exerts a wealth effect that reduces labor supply, further dampening production. Confronted with these forces, the planner relaxes the intertemporal budget constraint in the short run, allowing public borrowing to rise and using tax cuts to stimulate labor effort, capital accumulation, and ultimately the recovery of output. Lower taxes boost disposable income and consumption, but they also widen the budget deficit and temporarily increase the debt ratio, particularly because government spending remains elevated for several periods.

The contrasting behavior of fiscal variables between 2007 and 2021 thus reveals two distinct fiscal regimes. In 2007, the government had sufficient fiscal space to pursue a countercyclical debt policy—using borrowing and lower taxes to smooth the adjustment to higher spending—while in 2021, limited fiscal capacity constrained the planner’s choices, necessitating a more front-loaded fiscal consolidation. This divergence underscores the importance of initial conditions: high public indebtedness transforms the fiscal response from countercyclical to procyclical, amplifying the short-run output costs of stabilization.

In both scenarios, monetary policy plays a stabilizing role by anchoring inflation expectations. The planner uses interest-rate adjustments to maintain a near-constant inflation path, relying on fiscal tools to absorb real distortions. However, the effectiveness of this coordination depends crucially on the fiscal position. When debt is high, as in 2021, the burden of stabilization shifts toward monetary tightening and tax adjustments, whereas in 2007, with a lower debt ratio, the planner could rely on debt issuance as an additional buffer.

Comparing both scenarios also highlights how the introduction of monetary instruments alters the optimal debt dynamics relative to the fiscal-only environment. Without monetary tools, the planner’s response to a positive spending shock depends strongly on the persistence of the disturbance: transitory shocks are financed through temporary debt issuance, whereas persistent shocks require immediate fiscal adjustment. With monetary policy active, the planner gains an additional adjustment margin—short-term inflation control—which reduces the sensitivity of debt dynamics to persistence. In such a setting, even for persistent shocks, the planner may increase public debt temporarily to mitigate the decline in capital and consumption, while using the interest rate and inflation path to stabilize nominal variables. This mechanism is particularly relevant in the 2007 calibration, where moderate inflation and debt issuance jointly support aggregate demand. In contrast, under the 2021 calibration, the same mechanism is constrained by higher interest payments and fiscal rigidity, leading

the planner to substitute away from debt toward direct tax adjustments.

Overall, the comparison between 2007 and 2021 illustrates how monetary–fiscal interactions depend critically on the fiscal space and the persistence of shocks. When debt is low, the planner can exploit both instruments to smooth fluctuations and maintain welfare stability. When debt is high, monetary tightening complements fiscal consolidation, but at the cost of deeper short-term contractions in real activity. In both cases, maintaining stable inflation remains the central objective, yet the distribution of adjustment across policy instruments—and therefore across agents—varies sharply with the economy’s fiscal stance.

Summary. When monetary policy is active, inflation stabilization remains the priority, but fiscal instruments bear the main burden of debt adjustment. The higher the inherited debt ratio (as in 2021), the less fiscal space is available, requiring earlier and sharper tax increases to preserve solvency.

4.5 Model Dynamics under Fiscal Rules

Motivation. The stronger fiscal adjustments observed for the UK in 2021 relative to 2007 suggest that when fiscal tools are constrained or politically costly, monetary policy becomes the main channel of stabilization. In such cases, the economy may deviate from the Ramsey zero-inflation path, as taxpayers resist explicit tax hikes more than moderate inflation, which serves as an implicit and more diffuse fiscal instrument. Inflation thus operates as a “hidden tax” that reduces the real value of public debt, redistributes resources, and preserves fiscal space when direct tax adjustments are limited. Although it entails costs—such as price distortions and redistribution between savers and borrowers—moderate inflation can emerge as an endogenous response in high-debt or fiscally rigid economies. The following analysis examines this mechanism by exploring how inflation acts as an adjustment margin when fiscal instruments are restricted in the Ramsey framework (23)–(37).

Fiscal feedback rules. We now examine a setting in which fiscal instruments follow simple feedback rules rather than the Ramsey-optimal conditions. Specifically, each tax instrument $\Xi_t \in \{\kappa_t, \tau_t^k, \tau_t\}$ evolves according to:

$$\Xi_t = \Xi + \phi_\Xi(B_{t-1} - B), \quad (50)$$

where ϕ_Ξ governs the responsiveness of the fiscal instrument to deviations of debt from its steady-state level.

4.5.1 Spending shocks with fiscal rules

Upon examining Panels (a) of Figures 3 and 4 for the year 2021, it becomes evident that constraining the responsiveness of fiscal instruments significantly alters the transmission of government spending shocks. When tax parameters are prevented from adjusting optimally, the decline in private capital becomes markedly more pronounced. This contrasts with the baseline Ramsey scenario, in which fiscal policy in 2021 was allowed to react through sizeable changes in $\{\tau_t^k, \tau_t, \kappa_t\}$, mitigating the contraction in investment and output. The current restriction thus reveals a heightened need for alternative adjustment mechanisms once fiscal flexibility is lost.

The absence of tax adjustment generates a strong wealth effect on households. With higher government spending and static taxes, agents perceive a rise in their permanent income, prompting a reduction in labor supply. The resulting fall in hours worked amplifies the decline in output and triggers a mild recession. To preserve budget balance, the increase in government expenditure must be accommodated through higher borrowing, leading to a substantial rise in the public deficit. As fiscal instruments remain fixed, public debt becomes the primary variable absorbing the shock, climbing sharply before gradually returning to its steady state.

In this context, the social planner responds by lowering the nominal interest rate in an effort to stabilize economic activity. The monetary easing alleviates debt-servicing costs and temporarily supports aggregate demand. However, it also encourages additional borrowing, further increasing public debt in the short run. Debt peaks a few quarters after the initial shock, reflecting the cumulative impact of higher expenditures and lower revenues. To contain this debt buildup and ensure long-run fiscal sustainability, the planner subsequently reduces inflation, thereby raising the real burden of outstanding debt and anchoring expectations. This adjustment results in an optimal inflation path that departs from the zero-inflation benchmark, illustrating the dynamic interplay between monetary accommodation and debt stabilization.

Comparing the 2021 and 2007 calibrations underscores how fiscal capacity shapes this interaction. In 2021, the elevated debt-to-GDP ratio constrains fiscal space, forcing a more front-loaded reliance on monetary policy and inflation to restore equilibrium. By contrast, in 2007, fiscal instruments could adjust more freely, and the government possessed greater borrowing capacity to finance temporary spending increases. Consequently, the increase in public debt required to maintain budget equilibrium in 2021 is both larger and more persistent than in 2007, despite a similar magnitude of fiscal shock.

The comparison between Figures 3 and 4 highlights an additional insight: when fiscal parameters are held fixed, the persistence of the shock plays a limited role in shaping the qualitative trajectory of macroeconomic variables. This invariance arises because the lack of fiscal flexibility forces monetary policy to absorb most of the adjustment, regardless of

whether the shock is short- or long-lived. The muted fiscal response magnifies the wealth effect, dampening labor supply and accentuating the downturn in real activity. In contrast, when taxes adjust optimally, fiscal instruments serve as a stabilizing device, redistributing the burden of adjustment across agents and mitigating the impact of spending shocks on output and capital.

Overall, this exercise illustrates the critical role of fiscal tools in macroeconomic stabilization. When fiscal rules are rigid or politically constrained, the burden of adjustment shifts toward monetary policy and inflation dynamics. The resulting policy mix can maintain nominal stability but at the cost of deeper real contractions and slower debt convergence. These findings suggest that under high-debt conditions such as those of the UK in 2021, limited fiscal space not only constrains policy flexibility but also reshapes the transmission of shocks by forcing inflation to act as a residual adjustment mechanism.

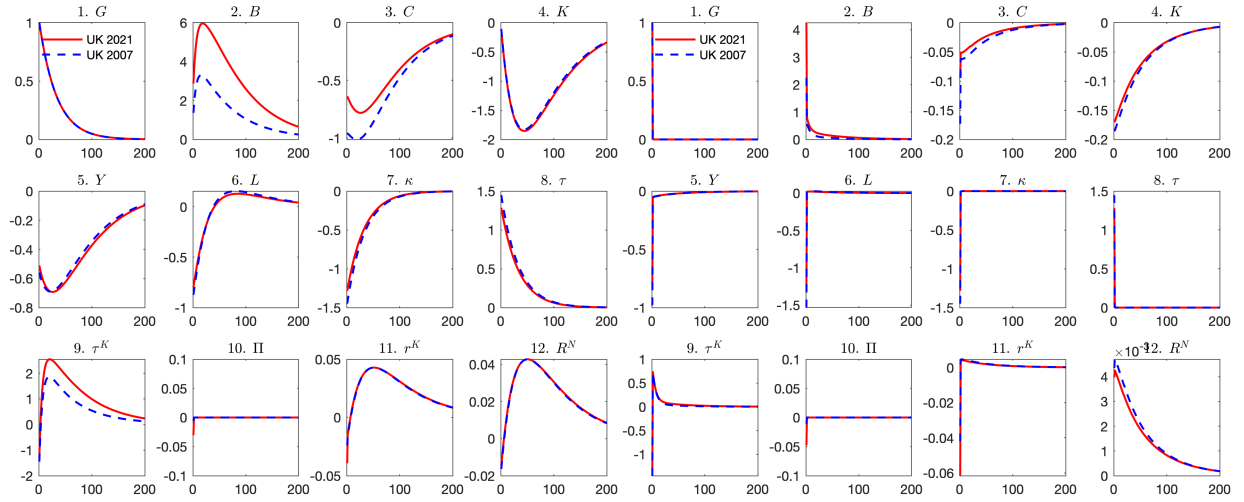


Figure 4: Positive government spending shock under fiscal feedback rules (50). UK 2021 (red solid) and UK 2007 (blue dashed).

4.5.2 Productivity shocks with fiscal rules

Figure 5 presents the responses to a negative productivity shock. The fall in output and capital is sharper than under the spending shock, resulting in a stronger debt response. The planner again relies on a combination of debt issuance and lower inflation to re-establish solvency. Inflation falls modestly below target, but the disinflation is more pronounced in 2021, consistent with a greater need for real fiscal adjustment.

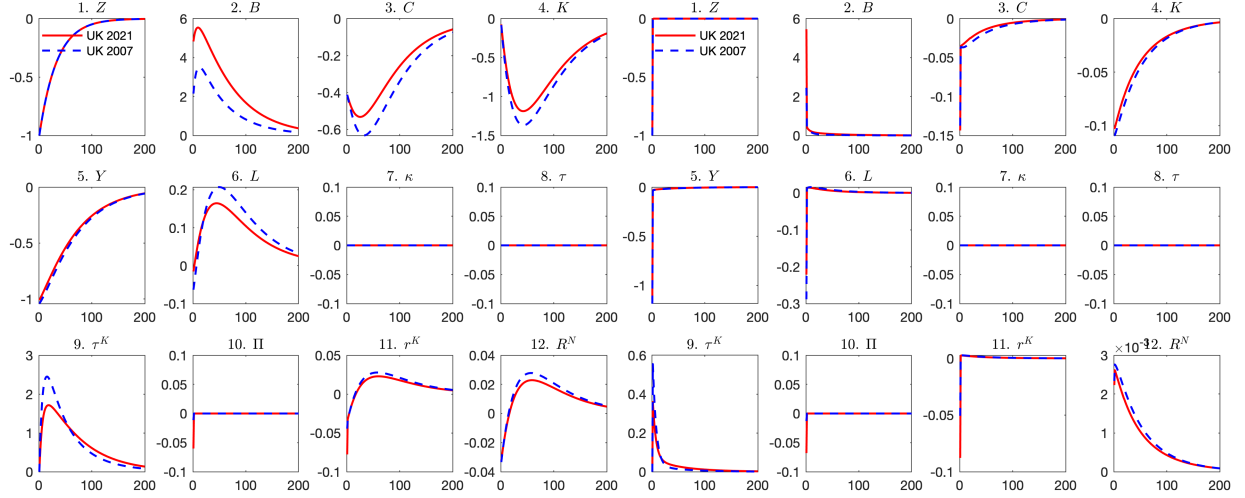


Figure 5: Negative productivity shock under fiscal feedback rules. UK 2021 (red solid) and UK 2007 (blue dashed).

Summary. When taxes adjust mechanically rather than optimally, inflation becomes an implicit state-contingent instrument. Even small movements in prices contribute to rebalancing the government’s budget constraint, particularly when the economy begins with high public debt. The larger magnitude of these responses in 2021 underscores the diminished fiscal capacity relative to 2007.

4.6 Model Dynamics under a Monetary Rule

Motivation. The results obtained thus far underscore the pivotal role of prices as an adjustment mechanism from an optimal policy perspective, particularly in settings where fiscal tools cannot be freely optimized but may adjust endogenously over the business cycle. This insight is especially relevant in the current macroeconomic environment, characterized by historically high public debt levels, where optimal fiscal policy would typically call for substantial tax adjustments to accommodate shocks. However, as discussed in Section 4.5, such adjustments may be politically or institutionally constrained.

Alternatively, one can interpret the problem from the standpoint of a monetary authority—such as the central bank—that lacks direct control over fiscal instruments but retains discretion over inflation and the nominal interest rate. To analyze this configuration, we solve the Ramsey problem (23)–(37) while closing the first-order conditions (FOCs) corresponding to fiscal policy instruments, given by equations (62)–(64). Fiscal policy parameters are thus held constant at their steady-state values, $(\tau_t^k, \tau_t, \kappa_t) = (\tau^k, \tau, \kappa)$, as reported in Table 2.

Policy rule. Within this framework, we introduce an active monetary authority following a simple feedback Taylor-type rule of the form:

$$\tilde{R}_t^n = R^n + \phi_{\Pi,t}(\Pi_t - 1), \quad (51)$$

where $\phi_{\Pi,t}$ determines the strength of the policy response to deviations of inflation from its target.

4.6.1 Spending shocks with a monetary rule

Figure 6 plots the responses of public debt B_t , consumption C_t , capital K_t , output Y_t , labor L_t , inflation Π_t , and the nominal interest rate $\tilde{R}_t^n$ following a positive government spending shock under the monetary policy rule (51).

The observed dynamics resemble those in Figure 2, where a positive spending shock induces a contraction in private capital across both calibrations. However, when fiscal policy instruments are fixed at their steady-state values, the adjustment mechanism shifts toward monetary policy and inflation. The planner (or monetary authority) tolerates a temporary deviation of inflation from target, using it as a stabilization tool to offset the inability to adjust taxes or debt optimally. When the shock persistence is low, the rise in inflation is modest (around 0.2%) and short-lived. Inflation increases slightly to facilitate consumption smoothing, allowing public debt to rise temporarily before gradually converging back to steady state. The mild inflationary response reduces real debt servicing costs without jeopardizing long-run price stability. In contrast, when the shock persistence is high, inflation rises much more sharply—by roughly 4%—to accommodate the stronger and more prolonged fiscal expansion. The initial increase in prices effectively lowers the real value of outstanding public debt, enabling the government to absorb the shock without immediate tax hikes. Public debt thus declines initially as inflation erodes its real burden, before increasing again as inflation is gradually reduced and the debt stock stabilizes. This “inflate-then-refinance” pattern mirrors the optimal response of a high-debt economy facing limited fiscal flexibility: inflation provides short-term relief from debt overhang but is swiftly corrected to prevent credibility losses.

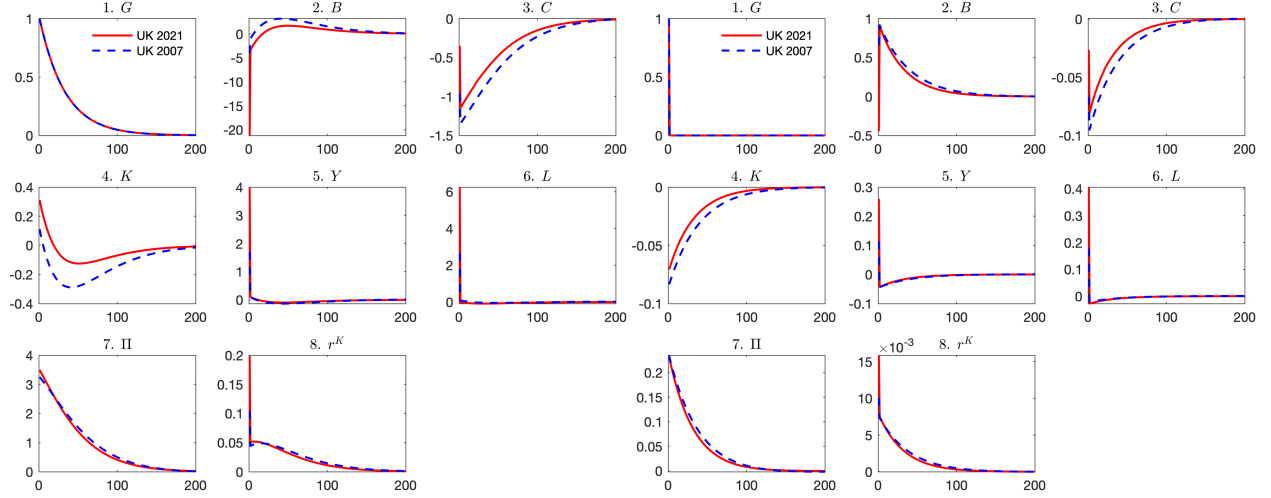


Figure 6: Positive government spending shock under a monetary policy rule (51). UK 2021 (red solid) and UK 2007 (blue dashed).

In summary, following a positive fiscal shock, inflation emerges as an effective adjustment instrument when fiscal tools are rigid. By reducing the real value of outstanding liabilities, moderate inflation temporarily relaxes fiscal constraints, mitigates debt accumulation, and supports aggregate demand. The planner's decision to allow short-term inflation can thus be viewed as an endogenous and optimal response to limited fiscal flexibility rather than as a deviation from policy discipline.

4.6.2 Productivity shocks with a monetary rule

We next consider the effects of a negative productivity shock under the same monetary policy rule. Figure 7 depicts the dynamic responses of macroeconomic aggregates for both the 2007 and 2021 calibrations, again distinguishing between high- and low-persistence shocks.

When the shock is highly persistent ($\rho_g = 0.97$), the monetary authority reacts by reducing the nominal interest rate to cushion the decline in output and investment. The resulting monetary expansion generates a temporary increase in inflation, which in turn lowers the real debt burden. This easing of debt-service costs allows the government to maintain solvency without raising taxes or cutting expenditures. The higher inflation also stimulates labor supply through a wealth effect, leading to a modest short-term recovery in output. Over time, as productivity and capital accumulation recover, inflation gradually subsides and the economy converges back to its steady state. In the low-persistence case, the negative productivity shock induces an immediate fall in capital, which dominates the short-run dynamics. The monetary authority again lowers the nominal interest rate, but the response is more moderate since the shock is short-lived. Inflation rises slightly, alleviating debt-service pressures but without triggering a substantial real adjustment. Public debt

increases modestly to finance temporary shortfalls in revenue, before stabilizing as inflation returns to target.

A key insight emerging from this exercise is that inflation plays a stabilizing role even in the face of real (supply-side) disturbances. When fiscal tools are fixed, the central bank uses inflation strategically to smooth consumption and sustain debt dynamics. The magnitude of the inflation response depends both on the persistence of the shock and the initial level of indebtedness: economies with higher public debt, such as the UK in 2021, exhibit stronger inflationary reactions because inflation acts as a substitute for unavailable fiscal instruments.

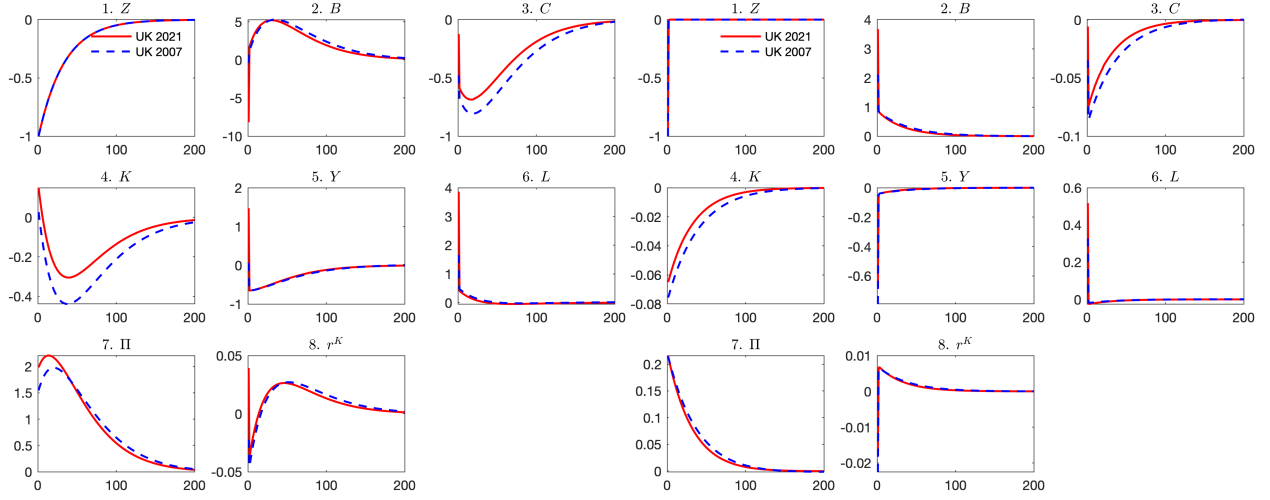


Figure 7: Negative productivity shock under a monetary policy rule (51). UK 2021 (red solid) and UK 2007 (blue dashed).

Summary. Across both fiscal and productivity shocks, the presence of an active monetary authority fundamentally changes the adjustment dynamics. When fiscal policy is constrained, inflation becomes the principal instrument for restoring equilibrium. The amplitude of the inflation response scales with both the persistence of the shock and the level of public indebtedness. High-debt economies exhibit stronger inflationary reactions, using price-level adjustments as a fiscal stabilization tool. This highlights the dual nature of inflation—as both a nominal anchor and an implicit fiscal instrument—when traditional tax-based stabilization mechanisms are unavailable.

4.7 Discussion

The results across all experiments highlight three key mechanisms governing the economy's adjustment dynamics. First, the initial level of public debt plays a central role. A higher starting debt ratio, as in 2021, tightens the intertemporal budget constraint and limits the scope for countercyclical fiscal policy. This reduced fiscal space forces the planner to rely

more heavily on tax adjustments and, when necessary, on inflation as a residual stabilization tool.

Second, the persistence of shocks critically influences policy responses. While transitory shocks can be absorbed through temporary borrowing and gradual fiscal adjustments, persistent disturbances require earlier consolidation and stronger fiscal reactions. The planner’s capacity to smooth consumption and investment over time is therefore closely tied to the duration of the shock.

Third, the interaction between fiscal and monetary policies shapes the composition of adjustment. When monetary instruments are available or fiscal rules restrict policy flexibility, inflation emerges as an implicit fiscal instrument. Allowing moderate, short-lived inflation can improve welfare by reducing the real debt burden and mitigating output losses, particularly in high-debt environments.

Overall, the findings underscore the importance of effective fiscal–monetary coordination. In 2021, with limited fiscal space, the UK economy relied more on price-level adjustments to absorb shocks. By contrast, in 2007, a lower debt burden permitted more active fiscal stabilization, smoother tax and debt dynamics, and stable inflation.

5 Concluding remarks

The post-COVID-19 and post-energy crisis period has highlighted the vital role of fiscal and monetary policy in stabilizing economies and supporting recovery. Using a calibrated model of the United Kingdom for 2007 and 2021—two periods with markedly different debt-to-GDP ratios—this paper examined how fiscal and monetary interactions shape macroeconomic adjustment.

Two main insights emerge. First, high public debt fundamentally constrains fiscal flexibility. When debt is low, as in 2007, the planner can use borrowing and tax adjustments to smooth consumption and output. When debt is high, as in 2021, limited fiscal space forces sharper, front-loaded consolidations and increases the reliance on monetary policy. Second, inflation becomes an endogenous adjustment mechanism when fiscal tools are constrained. Moderate, temporary inflation reduces the real debt burden and supports stabilization, effectively acting as a fiscal instrument when explicit taxation is limited.

These findings imply that low debt expands the set of available stabilization tools, while high debt shifts the adjustment burden toward prices. In such environments, moderate inflation can be a pragmatic, short-term response—provided it remains temporary and credible. Sustained macroeconomic stability thus hinges on close coordination between fiscal and monetary authorities.

During the preparation of this work the author used ChatGPT (OpenAI) in order to assist with grammar and language editing. After using this tool/service, the author reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix

A Derivation of the Government Budget Constraint

Substituting (11) into (12) gives:

$$G_t + \frac{\tilde{R}_{t-1}^n}{\Pi_t} B_{t-1} = D_t + \tilde{w}_t L_t - \kappa_t \int_i (\tilde{w}_t y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \tau_t^k \tilde{r}_t^k K_{t-1} + \tau_t^k \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right) B_{t-1} + B_t,$$

where $L_t = \int_i y_{i,t} l_{i,t} \ell(di)$ and D_t denotes firms' aggregate real profits, given by (8). Using $Y_t \xi_t = (\tilde{r}_t^k + \delta) K_{t-1} + \tilde{w}_t L_t$, we have:

$$D_t = \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t - \xi_t Y_t = \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t - [(\tilde{r}_t^k + \delta) K_{t-1} + \tilde{w}_t L_t].$$

Substituting this expression into the budget constraint yields:

$$\begin{aligned} G_t + \frac{\tilde{R}_{t-1}^n}{\Pi_t} B_{t-1} &= \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t - [(\tilde{r}_t^k + \delta) K_{t-1} + \tilde{w}_t L_t] + \tilde{w}_t L_t \\ &\quad - \kappa_t \int_i (\tilde{w}_t y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) + \tau_t^k \tilde{r}_t^k K_{t-1} + \tau_t^k \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right) B_{t-1} + B_t. \end{aligned}$$

We next define post-tax rates for capital, bonds, and labor income:

$$r_t = (1 - \tau_t^k) \tilde{r}_t, \quad r_t^k = (1 - \tau_t^k) \tilde{r}_t^k, \quad \frac{R_t^n}{\Pi_t} = (1 - \tau_t^k) \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right), \quad \text{and} \quad w_t = \kappa_t (\tilde{w}_t)^{1-\tau_t}.$$

Using these definitions, the government budget constraint becomes:

$$G_t + \frac{R_t^n}{\Pi_t} B_{t-1} + B_{t-1} + r_t^k K_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t - \delta K_{t-1} + B_t.$$

Finally, using (9), we have:

$$(1 - \tau_t^k) \tilde{r}_t A_{t-1} = (1 - \tau_t^k) \tilde{r}_t^k K_{t-1} + (1 - \tau_t^k) \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right) B_{t-1},$$

which, together with the definitions in (13), implies $r_t A_{t-1} = r_t^k K_{t-1} + \frac{R_t^n}{\Pi_t} B_{t-1}$. The final government budget constraint can therefore be written as:

$$G_t + B_{t-1} + r_t A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) = \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t - \delta K_{t-1} + B_t.$$

B The Lagrangian and First Order Conditions of the Ramsey Problem

The Ramsey problem (23)–(37) can be rewritten as:

$$\begin{aligned}
\mathcal{L} = & \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \omega_{i,t} (u(c_{i,t}) - v(l_{i,t})) \ell(di) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i (\lambda_{i,c,t} - (1+r_t) \lambda_{i,c,t-1}) u'(c_{i,t}) \ell(di) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \int_i \lambda_{i,l,t} (v'(l_{i,t}) - (1-\tau_t) w_t y_{i,t} (y_{i,t} l_{i,t})^{-\tau_t} u'(c_{i,t})) \ell(di) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t (\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) Y_t M_t \\
& + \frac{\epsilon - 1}{\psi} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \gamma_t (\xi_t - 1) Y_t M_t \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \mu_t \left(G_t + (1-\delta) B_{t-1} + (r_t + \delta) A_{t-1} + w_t \int_i (y_{i,t} l_{i,t})^{1-\tau_t} \ell(di) - \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) Y_t - B_t \right) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Gamma_t \left((1 - \tau_t^k) \tilde{r}_t^k K_{t-1} + (1 - \tau_t^k) \left(\frac{\tilde{R}_{t-1}^n}{\Pi_t} - 1 \right) B_{t-1} - r_t A_{t-1} \right) \\
& - \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \Psi_t \left(\mathbb{E}_t[1 + \tilde{r}_{t+1}^k] - \mathbb{E}_t \left[\frac{\tilde{R}_t^n}{\Pi_{t+1}} \right] \right).
\end{aligned} \tag{52}$$

The instruments are $\Pi_t, \tilde{R}_t^n, \tilde{r}_t^k, a_{i,t}, B_t, l_{i,t}, w_t, r_t$, and τ_t .

FOC with respect to Π_t . Differentiating (52) with respect to Π_t yields:

$$0 = -(\gamma_t - \gamma_{t-1})(2\Pi_t - 1)Y_t M_t - \mu_t \psi (\Pi_t - 1)Y_t - \Gamma_t (1 - \tau_t^k) B_{t-1} \frac{-\tilde{R}_{t-1}^n}{\Pi_t^2} + \Psi_{t-1} \beta^{-1} \frac{-\tilde{R}_{t-1}^n}{\Pi_t^2}.$$

Rearranging gives:

$$\begin{array}{lll}
\underbrace{\mu_t \psi (\Pi_t - 1)}_{\text{Total cost due to inflation}} & = & \underbrace{(\gamma_{t-1} - \gamma_t)(2\Pi_t - 1)M_t}_{\text{Manipulation of real wages via the Phillips curve}} + \underbrace{\left(\Gamma_t (1 - \tau_t^k) B_{t-1} - \beta^{-1} \Psi_{t-1} \right) \frac{\tilde{R}_{t-1}^n}{\Pi_t^2 Y_t}}_{\text{Reduction of real interest payments on debt}}.
\end{array}$$

Hence,

$$0 = \mu_t \psi (\Pi_t - 1) + (\gamma_t - \gamma_{t-1})(2\Pi_t - 1)M_t - \left(\Gamma_t (1 - \tau_t^k) B_{t-1} - \beta^{-1} \Psi_{t-1} \right) \frac{\tilde{R}_{t-1}^n}{\Pi_t^2 Y_t}. \tag{53}$$

FOC with respect to $\tilde{R}_t^n$. Differentiating (52) with respect to $\tilde{R}_t^n$ yields:

$$0 = -\beta\Gamma_{t+1}(1 - \tau_{t+1}^k)\frac{B_t}{\Pi_{t+1}} + \Psi_t\frac{1}{\Pi_{t+1}}. \quad (54)$$

FOC with respect to $\tilde{r}_t^k$. Differentiating (52) with respect to $\tilde{r}_t^k$ yields:

$$-\Gamma_t(1 - \tau_t^k)K_{t-1} + \left(\frac{\epsilon - 1}{\psi}\right)\gamma_t\frac{1}{\alpha}K_{t-1}M_t - \beta^{-1}\Psi_{t-1} = 0.$$

Thus,

$$\beta^{-1}\Psi_{t-1} = \left(\left(\frac{\epsilon - 1}{\psi}\right)\gamma_t\frac{1}{\alpha}M_t - \Gamma_t(1 - \tau_t^k)\right)K_{t-1}. \quad (55)$$

FOC with respect to $a_{i,t}$. Differentiating (52) with respect to $a_{i,t}$ yields:

$$\begin{aligned}
0 = & \beta^t \int_j \omega_{j,t} u'(c_{j,t}) \frac{\partial c_{j,t}}{\partial a_{i,t}} \ell(dj) \\
& - \beta^t \int_j (\lambda_{j,c,t} - (1 + r_t) \lambda_{j,c,t-1}) u''(c_{j,t}) \frac{\partial c_{j,t}}{\partial a_{i,t}} \ell(dj) \\
& + \beta^t (1 - \tau_t) w_t \int_j \lambda_{j,l,t} (y_{j,t})^{1-\tau_t} (l_{j,t})^{-\tau_t} u''(c_{j,t}) \frac{\partial c_{j,t}}{\partial a_{i,t}} \ell(dj) \\
& - \beta^t \left((\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) Y_t - \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t (\xi_t - 1) Y_t \right) \int_j u''(c_{j,t}) \frac{\partial c_{j,t}}{\partial a_{i,t}} \ell(dj) \\
& + \beta^t \mu_t \left(\left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) \frac{\partial Y_t}{\partial a_{i,t}} - (r_t + \delta) \frac{\partial A_{t-1}}{\partial a_{i,t}} \right) \\
& + \beta^{t+1} \mathbb{E}_t \left[\int_j \omega_{j,t+1} u'(c_{j,t+1}) \frac{\partial c_{j,t+1}}{\partial a_{i,t}} \ell(dj) \right] \\
& - \beta^{t+1} \mathbb{E}_t \left[\int_j (\lambda_{j,c,t+1} - (1 + r_{t+1}) \lambda_{j,c,t}) u''(c_{j,t+1}) \frac{\partial c_{j,t+1}}{\partial a_{i,t}} \ell(dj) \right] \\
& + \beta^{t+1} (1 - \tau_{t+1}) w_{t+1} \mathbb{E}_t \left[\int_j \lambda_{j,l,t+1} (y_{j,t+1})^{1-\tau_{t+1}} (l_{j,t+1})^{-\tau_{t+1}} u''(c_{j,t+1}) \frac{\partial c_{j,t+1}}{\partial a_{i,t}} \ell(dj) \right] \\
& - \beta^{t+1} \mathbb{E}_t \left[\left((\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) Y_{t+1} - \left(\frac{\epsilon - 1}{\psi} \right) \gamma_{t+1} (\xi_{t+1} - 1) Y_{t+1} \right) \int_j u''(c_{j,t+1}) \frac{\partial c_{j,t+1}}{\partial a_{i,t+1}} \ell(dj) \right] \\
& + \beta^{t+1} \mathbb{E}_t \left[\mu_{t+1} \left(\left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \frac{\partial Y_{t+1}}{\partial a_{i,t}} - (r_{t+1} + \delta) \frac{\partial A_t}{\partial a_{i,t}} \right) \right] \\
& - \beta^{t+1} \mathbb{E}_t \left[\left((\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) - \left(\frac{\epsilon - 1}{\psi} \right) \gamma_{t+1} (\xi_{t+1} - 1) \right) M_{t+1} \frac{\partial Y_{t+1}}{\partial a_{i,t}} \right] \\
& + \beta^{t+1} \mathbb{E}_t [\Gamma_{t+1} r_{t+1}] - \beta^{t+1} \mathbb{E}_t \left[\Gamma_{t+1} \left((1 - \tau_{t+1}^k) \tilde{r}_{t+1}^k \right) \right] \\
& + \underbrace{\left(\frac{\epsilon - 1}{\psi} \right) \beta^{t+1} \mathbb{E}_t \left[\gamma_{t+1} Y_{t+1} M_{t+1} \frac{\tilde{r}_{t+1}^k + \delta (1 - \alpha)}{\alpha} \frac{1}{Y_{t+1}} \right]}_{\left(\frac{\epsilon - 1}{\psi} \right) \beta^{t+1} \mathbb{E}_t \left[\gamma_{t+1} Y_{t+1} M_{t+1} \xi_{t+1} (1 - \alpha) \frac{1}{K_t} \right]}.
\end{aligned}$$

Observe that:

$$\begin{aligned}
& \left(\frac{\epsilon - 1}{\psi} \right) \beta^{t+1} \mathbb{E}_t \left[\gamma_{t+1} (\xi_{t+1} - 1) M_{t+1} \alpha K_t^{\alpha-1} L_{t+1}^{1-\alpha} \frac{K_t}{K_t} \right] + \left(\frac{\epsilon - 1}{\psi} \right) \beta^{t+1} \mathbb{E}_t \left[\gamma_{t+1} Y_{t+1} M_{t+1} \xi_{t+1} (1 - \alpha) \frac{1}{K_t} \right] \\
& = \left(\frac{\epsilon - 1}{\psi} \right) \beta^{t+1} \mathbb{E}_t \left[\gamma_{t+1} (\xi_{t+1} - \alpha) M_{t+1} \frac{Y_{t+1}}{K_t} \right].
\end{aligned}$$

Also note that we use $K_{t-1} = A_{t-1} - B_{t-1}$ to simplify the FOCs. The last term above arises from $\xi_t = \left(\frac{\tilde{r}_t^k + \delta}{\alpha} \right) \frac{K_{t-1}}{Y_t}$, so $\frac{\partial \xi_{t+1}}{\partial a_{i,t}} = (1 - \alpha) \frac{\tilde{r}_{t+1}^k + \delta}{\alpha} \frac{1}{Y_{t+1}}$.

Define:

$$\begin{aligned} \psi_{i,t} = & \omega_{i,t} u'(c_{i,t}) - \left[\lambda_{i,c,t} - (1 + r_t) \lambda_{i,c,t-1} - \lambda_{i,l,t} (1 - \tau_t) w_t(y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \right. \\ & \left. + \left((\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) - \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t (\xi_t - 1) \right) Y_t \right] u''(c_{i,t}). \end{aligned} \quad (56)$$

Using (25) and (37), we obtain:

$$\begin{aligned} \frac{\partial c_{j,t}}{\partial a_{i,t}} &= -1_{i=j}, & \frac{\partial A_t}{\partial a_{i,t}} &= 1, \\ \frac{\partial c_{j,t+1}}{\partial a_{i,t}} &= (1 + r_{t+1}) 1_{i=j}, & \frac{\partial Y_{t+1}}{\partial a_{i,t}} &= \alpha K_t^{\alpha-1} L_{t+1}^{1-\alpha}. \end{aligned}$$

Hence,

$$\begin{aligned} \psi_{i,t} = & \beta \mathbb{E}_t [(1 + r_{t+1}) \psi_{i,t+1}] + \beta \mathbb{E}_t \left[\mu_{t+1} \left(\left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \underbrace{\alpha K_t^{\alpha-1} L_{t+1}^{1-\alpha}}_{\left(\frac{\tilde{r}_{t+1}^k + \delta}{\xi_{t+1}} \right)} - r_{t+1} - \delta \right) \right] \\ & + \left(\frac{\epsilon - 1}{\psi} \right) \beta \mathbb{E}_t \left[\gamma_{t+1} (\xi_{t+1} - \alpha) M_{t+1} \frac{Y_{t+1}}{K_t} \right] - \beta \mathbb{E}_t \left[(\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \alpha \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ & - \beta \mathbb{E}_t [\Gamma_{t+1} (1 - \tau_{t+1}^k) \tilde{r}_{t+1}^k] + \beta \mathbb{E}_t [\Gamma_{t+1} r_{t+1}]. \end{aligned} \quad (57)$$

FOC with respect to B_t . Before deriving the first-order condition with respect to B_t , we substitute $K_{t-1} = A_{t-1} - B_{t-1}$. Differentiating (52) with respect to B_t yields:

$$\begin{aligned} 0 = & \mu_t - \beta \mathbb{E}_t \left[\mu_{t+1} \left(1 - \delta + \left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \underbrace{\alpha K_t^{\alpha-1} L_{t+1}^{1-\alpha}}_{\left(\frac{\tilde{r}_{t+1}^k + \delta}{\xi_{t+1}} \right)} \right) \right] \\ & - \left(\frac{\epsilon - 1}{\psi} \right) \beta \mathbb{E}_t \left[\gamma_{t+1} (\xi_{t+1} - \alpha) M_{t+1} \frac{Y_{t+1}}{K_t} \right] + \beta \mathbb{E}_t \left[(\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \alpha \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ & - \beta \mathbb{E}_t \left[\Gamma_{t+1} (1 - \tau_{t+1}^k) \left(\frac{\tilde{R}_t^n}{\Pi_{t+1}} - 1 \right) \right] + \beta \mathbb{E}_t [\Gamma_{t+1} (1 - \tau_{t+1}^k) \tilde{r}_{t+1}^k]. \end{aligned}$$

Thus,

$$\begin{aligned}\mu_t = & \beta \mathbb{E}_t \left[\mu_{t+1} \left(1 - \delta + \left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \left(\frac{\tilde{r}_{t+1}^k + \delta}{\xi_{t+1}} \right) \right) \right] \\ & + \beta \mathbb{E}_t \left[\left(\left(\frac{\epsilon - 1}{\psi} \right) \gamma_{t+1} (\xi_{t+1} - \alpha) - \alpha (\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \right) \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ & + \beta \mathbb{E}_t \left[\Gamma_{t+1} (1 - \tau_{t+1}^k) \left(\frac{\tilde{R}_t^n}{\Pi_{t+1}} - 1 - \tilde{r}_{t+1}^k \right) \right].\end{aligned}\quad (58)$$

Combining (57)–(58) we have:

$$\begin{aligned}\psi_{i,t} - \mu_t = & \beta \mathbb{E}_t [\psi_{i,t} (1 + r_{t+1})] - \beta \mathbb{E}_t [\mu_{t+1} (1 - \delta)] + \beta \mathbb{E}_t [\mu_{t+1} (-r_{t+1} - \delta)] - \beta \mathbb{E}_t [\Gamma_{t+1} (1 - \tau_{t+1}^k) \tilde{r}_{t+1}^k] \\ & + \beta \mathbb{E}_t [\Gamma_{t+1} r_{t+1}] - \beta \mathbb{E}_t \left[\Gamma_{t+1} (1 - \tau_{t+1}^k) \left(\frac{\tilde{R}_t^n}{\Pi_{t+1}} - 1 \right) \right] + \beta \mathbb{E}_t [\Gamma_{t+1} (1 - \tau_{t+1}^k) \tilde{r}_{t+1}^k].\end{aligned}$$

Defining:

$$\hat{\psi}_{i,t} = \psi_{i,t} - \mu_t, \quad (59)$$

we obtain:

$$\hat{\psi}_{i,t} = \beta \mathbb{E}_t [\hat{\psi}_{i,t} (1 + r_{t+1})] + \beta \mathbb{E}_t \left[\Gamma_{t+1} \left(r_{t+1} - (1 - \tau_{t+1}^k) \left(\frac{\tilde{R}_t^n}{\Pi_{t+1}} - 1 \right) \right) \right]. \quad (60)$$

FOC with respect to $l_{i,t}$. Differentiating (52) with respect to $l_{i,t}$ yields:

$$\begin{aligned}0 = & \int_j \psi_{j,t} \frac{\partial c_{j,t}}{\partial l_{i,t}} \ell(dj) - \omega_{i,t} v'(l_{i,t}) - \lambda_{i,t} v''(l_{i,t}) \\ & + \lambda_{i,t} (1 - \tau_t) (-\tau_t) w_t(y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t-1} u'(c_{i,t}) - \mu_t \left(w_t (1 - \tau_t) (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \right. \\ & \left. - \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) \frac{\partial Y_t}{\partial l_{i,t}} \right) - (\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) M_t \frac{\partial Y_t}{\partial l_{i,t}} + \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t (\xi_t - 1) M_t \frac{\partial Y_t}{\partial l_{i,t}} \\ & + \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t Y_t M_t \underbrace{\alpha \frac{\tilde{w}_t}{1 - \alpha} \frac{1}{Y_t} y_{i,t}}_{\frac{\partial \xi_t}{\partial l_{i,t}}} \\ & \underbrace{\left(\frac{\epsilon - 1}{\psi} \right) \gamma_t \alpha M_t \frac{\xi_t Y_t}{L_t} y_{i,t}}_{\left(\frac{\epsilon - 1}{\psi} \right) \gamma_t \alpha M_t \frac{\xi_t Y_t}{L_t} y_{i,t}}.\end{aligned}$$

Note that $\xi_t = \left(\frac{\tilde{w}_t}{1 - \alpha} \right) \frac{L_t}{Y_t}$ and $L_t = \int_i y_{i,t} l_{i,t} \ell(di)$. Therefore, $\frac{\partial \xi_t}{\partial l_{i,t}} = \alpha \frac{\tilde{w}_t}{1 - \alpha} \frac{1}{Y_t} y_{i,t}$. Using (25) and (37) we obtain, respectively, $\frac{\partial c_{j,t}}{\partial l_{i,t}} = (1 - \tau_t) w_t(y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} 1_{i=j}$ and $\frac{\partial Y_t}{\partial l_{i,t}} = (1 -$

$\alpha) K_{t-1}^\alpha L_t^{-\alpha} y_{i,t} = F_{L,t} y_{i,t} = \frac{\tilde{w}_t}{\xi_t} y_{i,t}$. So:

$$\begin{aligned} \omega_{i,t} v'(l_{i,t}) + \lambda_{i,l,t} v''(l_{i,t}) &= (1 - \tau_t) w_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t} \hat{\psi}_{i,t} \\ &\quad + \lambda_{i,l,t} (1 - \tau_t) (-\tau_t) w_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t-1} u'(c_{i,t}) + \mu_t \left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) F_{L,t} y_{i,t} \\ &\quad - (\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) M_t \frac{(1 - \alpha) Y_t y_{i,t}}{L_t} \\ &\quad + \underbrace{\left(\frac{\epsilon - 1}{\psi} \right) \gamma_t (\xi_t - 1) M_t \frac{(1 - \alpha) Y_t y_{i,t}}{L_t} + \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t \xi_t M_t \frac{\alpha Y_t y_{i,t}}{L_t}}_{\left(\frac{\epsilon - 1}{\psi} \right) \gamma_t (\xi_t - (1 - \alpha)) M_t \frac{Y_t y_{i,t}}{L_t}}. \end{aligned}$$

where $\hat{\psi}_t^i = \psi_t^i - \mu_t$. After some manipulation we obtain:

$$\begin{aligned} \frac{\omega_{i,t} v'(l_{i,t}) + \lambda_{i,l,t} v''(l_{i,t})}{(1 - \tau_t) w_t (y_{i,t})^{1-\tau_t} (l_{i,t})^{-\tau_t}} &= \hat{\psi}_{i,t} - \lambda_{i,l,t} \tau_t \frac{u'(c_{i,t})}{l_{i,t}} + \mu_t \frac{\left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) F_{L,t}}{(1 - \tau_t) w_t (y_{i,t})^{-\tau_t} (l_{i,t})^{-\tau_t}} \\ &\quad + \left[\left(\frac{\epsilon - 1}{\psi} \right) \gamma_t (\xi_t - (1 - \alpha)) - (1 - \alpha) (\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) \right] \frac{Y_t M_t}{L_t (1 - \tau_t) w_t (y_{i,t})^{-\tau_t} (l_{i,t})^{-\tau_t}}. \end{aligned} \quad (61)$$

FOC with respect to w_t . Differentiating (52) with respect to w_t yields:

$$\begin{aligned} 0 &= \int_j \left(\psi_{j,t} \frac{\partial c_{j,t}}{\partial w_t} + \lambda_{j,l,t} (1 - \tau_t) (y_{j,t})^{1-\tau_t} (l_{j,t})^{-\tau_t} u'(c_{j,t}) \right) \ell(dj) \\ &\quad - \mu_t \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) + \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t Y_t M_t \frac{\partial \xi_t}{\partial w_t}. \end{aligned}$$

Using (25), $\frac{\partial c_{j,t}}{\partial w_t} = (y_{j,t} l_{j,t})^{1-\tau_t}$. Since $w_t = \kappa (\tilde{w}_t)^{1-\tau_t}$, we have $\tilde{w}_t = \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}}$. Hence $\xi_t = \left(\frac{\left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}}}{1 - \alpha} \right) \frac{L_t}{Y_t}$, so:

$$\frac{\partial \xi_t}{\partial w_t} = \frac{\frac{1}{1-\tau_t} \left(\frac{w_t}{\kappa} \right)^{\frac{\tau_t}{1-\tau_t}} \frac{1}{\kappa} \frac{L_t}{Y_t}}{1 - \alpha}.$$

Substituting back, we obtain:

$$\begin{aligned} 0 &= \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{j,l,t} (1 - \tau_t) u'(c_{j,t}) / l_{j,t} \right) \ell(dj) \\ &\quad + \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t M_t \frac{1}{1 - \tau_t} \left(\frac{w_t}{\kappa} \right)^{\frac{\tau_t}{1-\tau_t}} \frac{1}{\kappa} \frac{L_t}{1 - \alpha}. \end{aligned} \quad (62)$$

FOC with respect to r_t . Differentiating (52) with respect to r_t yields:

$$0 = \int_j \left(\psi_{j,t} \frac{\partial c_{j,t}}{\partial r_t} + \lambda_{j,c,t-1} u'(c_{j,t}) \right) \ell(dj) - \mu_t \int_j a_{j,t-1} \ell(dj) + \Gamma_t (B_{t-1} + K_{t-1}).$$

Using (25), $\frac{\partial c_{j,t}}{\partial r_t} = a_{j,t-1}$. Hence,

$$0 = \int_j \left(\hat{\psi}_{j,t} a_{j,t-1} + \lambda_{j,c,t-1} u'(c_{j,t}) \right) \ell(dj) + \Gamma_t (B_{t-1} + K_{t-1}). \quad (63)$$

FOC with respect to τ_t . Differentiating (52) with respect to τ_t yields:

$$\begin{aligned} 0 = & \int_j \psi_{j,t} \frac{\partial c_{j,t}}{\partial \tau_t} \ell(dj) + w_t \int_j \lambda_{j,l,t} \frac{\partial}{\partial \tau_t} \left((1 - \tau_t) (y_{j,t} l_{j,t})^{1-\tau_t} \right) (u'(c_{j,t})/l_{j,t}) \ell(dj) \\ & - \mu_t w_t \int_j \frac{\partial}{\partial \tau_t} \left((y_{j,t} l_{j,t})^{1-\tau_t} \right) \ell(dj) + \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t Y_t M_t \frac{\partial \xi_t}{\partial \tau_t}. \end{aligned}$$

Using (25), $\frac{\partial c_{j,t}}{\partial \tau_t} = w_t \frac{\partial}{\partial \tau_t} \left((y_{j,t} l_{j,t})^{1-\tau_t} \right)$. Let $y = (y_{j,t} l_{j,t})^{1-\tau_t}$. Then $\ln y = (1 - \tau_t) \ln(y_{j,t} l_{j,t})$ and $\frac{\partial y}{\partial \tau_t} = -\ln(y_{j,t} l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t}$, so:

$$\frac{\partial}{\partial \tau_t} \left((y_{j,t} l_{j,t})^{1-\tau_t} \right) = -\ln(y_{j,t} l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t}.$$

Similarly, with $y = \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}}$, we have $\frac{\partial y}{\partial \tau_t} = \frac{1}{(1-\tau_t)^2} \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}} \ln \left(\frac{w_t}{\kappa} \right)$, implying:

$$\frac{\partial \xi_t}{\partial \tau_t} = \frac{1}{(1 - \tau_t)^2} \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}} \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{(1 - \alpha) Y_t}.$$

Therefore,

$$\begin{aligned} 0 = & \int_j \hat{\psi}_{j,t} w_t \ln(y_{j,t} l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) + w_t \int_j \lambda_{j,l,t} (1 - \tau_t) (u'(c_{j,t})/l_{j,t}) \ln(y_{j,t} l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) \\ & + w_t \int_j \lambda_{j,l,t} (u'(c_{j,t})/l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) - \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t Y_t M_t \frac{1}{(1 - \tau_t)^2} \left(\frac{w_t}{\kappa} \right)^{\frac{1}{1-\tau_t}} \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{(1 - \alpha) Y_t}. \end{aligned}$$

Equivalently,

$$\begin{aligned} 0 = & \int_j (y_{j,t} l_{j,t})^{1-\tau_t} \left(\hat{\psi}_{j,t} + \lambda_{j,l,t} (1 - \tau_t) (u'(c_{j,t})/l_{j,t}) \right) \ln(y_{j,t} l_{j,t}) \ell(dj) \\ & + \int_j \lambda_{j,l,t} (u'(c_{j,t})/l_{j,t}) (y_{j,t} l_{j,t})^{1-\tau_t} \ell(dj) - \left(\frac{\epsilon - 1}{\psi} \right) \gamma_t M_t \frac{1}{(1 - \tau_t)^2} \left(\frac{w_t^{\frac{\tau_t}{1-\tau_t}}}{\kappa^{\frac{1}{1-\tau_t}}} \right) \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{(1 - \alpha)}. \end{aligned} \quad (64)$$

C Truncation Method for the Ramsey Problem with Nominal Rigidities

This appendix develops a truncation framework for the Ramsey problem studied in the main text, which features nominal rigidities (Rotemberg adjustment), inflation Π_t , the New Keynesian Phillips block via the auxiliary objects (γ_t, ξ_t, M_t) , and fiscal/asset-pricing wedges. We construct a finite state-space (truncated) economy that (i) exactly aggregates the heterogeneous-agent model in steady state and (ii) delivers accurate local dynamics under small aggregate shocks, while preserving the planner's first-order conditions (FOCs) (53)–(64).

C.1 The truncated model

Truncated histories. The key step of the aggregation procedure consists in assigning the same wealth and allocation to all agents who share the same idiosyncratic history over the recent past. This recent past is characterized by a fixed number of periods, called the *truncation length* and denoted by N . The truncation length is a parameter of the model. The N -period history is referred to as a *truncated history*.

For an individual history $y^t = \{\dots, y_{t-N}^t, y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t\}$, the truncated version is the N -length vector:

$$y^N := \{y_{t-N}^t, y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t\}.$$

Thus, the truncated history of an agent i whose idiosyncratic history is y^t can be represented as:

$$y^t = \left\{ \underbrace{\dots, y_{t-N-2}^t, y_{t-N-1}^t, y_{t-N}^t}_{\xi_{y^N}}, \underbrace{y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t}_{=y^N} \right\},$$

where the parameter ξ_{y^N} captures the residual heterogeneity associated with the truncated history y^N . The variable y_{t-k}^t represents the idiosyncratic state k periods before date t . The procedure to compute the set of parameters $(\xi_{y^N})_{y^N}$ will be detailed below. In what follows, we outline the main elements needed to implement this aggregation process.

Population shares and aggregation of individual choices. First, we compute the measure of agents who share the same truncated history y^N . An agent with history $\tilde{y}^N$ at time $t-1$ will have a different truncated history at time t depending on the realization of the idiosyncratic state $y_{i,t}$. The probability of transitioning from truncated history $\tilde{y}^N$ to y^N is denoted by $\Pi_{\tilde{y}^N y^N}$, with $\sum_{y^N \in \mathcal{Y}^N} \Pi_{\tilde{y}^N y^N} = 1$. It can be derived from the transition

probabilities of the underlying idiosyncratic process as:

$$\Pi_{\tilde{y}^N y^N} = 1_{y^N \succeq \tilde{y}^N} \Pi_{\tilde{y}_0^N y_0^N} \geq 0,$$

where $1_{y^N \succeq \tilde{y}^N}$ equals one if y^N is a possible continuation of $\tilde{y}^N$ and zero otherwise. Using these elements, we can determine the share of agents with truncated history y^N at time t , denoted by S_{t,y^N} , which evolves according to:

$$S_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} S_{t-1,\tilde{y}^N} \Pi_{\tilde{y}^N y^N}, \quad (65)$$

where the initial shares $(S_{-1,y^N})_{y^N \in \mathcal{Y}^N}$ are given and satisfy $\sum_{y^N \in \mathcal{Y}^N} S_{-1,y^N} = 1$.

The aggregation of the model then assigns to each truncated history y^N the average of all individual choices—whether for consumption, savings, or labor supply—of the group of agents sharing that same truncated history. Consider a generic variable $X_t(y^t, s^t)$ and denote by X_{t,y^N} the average value of X assigned to the group with truncated history y^N . Formally:

$$X_{t,y^N} = \frac{1}{S_{t,y^N}} \sum_{y^t \in \mathcal{Y}^{t+1} | (y_{t-N+1}^t, \dots, y_{t-1}^t, y_t^t) = y^N} X_t(y^t, s^t) \mu_t(y^t), \quad (66)$$

where $\mu_t(y^t)$ denotes the measure of agents with history y^t .

Equation (66) can be applied to key individual-level variables such as consumption, savings, labor supply, and the Lagrange multiplier on the borrowing constraint. This leads to the aggregate quantities c_{t,y^N} , $\tilde{a}_{t,y^N}$, l_{t,y^N} , and ν_{t,y^N} , respectively.

When applying the aggregation procedure to beginning-of-period wealth, one must account for the fact that agents with truncated history y^N at time t may have originated from multiple truncated histories at $t - 1$. Specifically, the beginning-of-period wealth for truncated history y^N , denoted $\tilde{a}_{t,y^N}$, is the weighted average of wealth among all groups $\tilde{y}^N$ at $t - 1$ that can transition into y^N :

$$\tilde{a}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^N} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \quad (67)$$

This expression ensures consistency in the aggregation of beginning-of-period asset holdings across truncated histories.

Nonlinear aggregation coefficients. Because the planner's FOCs involve nonlinear functions (notably $u(\cdot)$, $v(\cdot)$, $u'(\cdot)$, $v'(\cdot)$, powers of (yl) , and cross-terms with $u''(\cdot)$), we introduce coefficients that map the sum of individual nonlinearities within each truncated group to functions of group-level averages. For each truncated history y^N , we define a set of steady-

state coefficients (ξ_{y^N}) satisfying the following relations:

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} u(c_{i,t}) = \xi_{y^N}^{u,0} u(c_{t,y^N}), \quad \sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} v(l_{i,t}) = \xi_{y^N}^{v,0} v(l_{t,y^N}), \quad (68)$$

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} u'(c_{i,t}) = \xi_{y^N}^{u,1} u'(c_{t,y^N}), \quad \sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} v'(l_{i,t}) = \xi_{y^N}^{v,1} v'(l_{t,y^N}), \quad (69)$$

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} (y_{i,t} l_{i,t})^{1-\tau_t} = \xi_{y^N}^y (y_0^N l_{t,y^N})^{1-\tau_t}, \quad \sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} \frac{u'(c_{i,t})}{l_{i,t}} = \xi_{y^N}^{u,\ell} \frac{u'(c_{t,y^N})}{l_{t,y^N}}. \quad (70)$$

Here, y_0^N denotes the most recent idiosyncratic component in the truncated history y^N .

For log-product terms appearing in the progressivity condition, we similarly define:

$$\sum_{y_i^t \in \mathcal{Y}^{t+1} | y_i^{t,N} = y^N} (y_{i,t} l_{i,t})^{1-\tau_t} \ln(y_{i,t} l_{i,t}) = \xi_{y^N}^{y \ln} (y_0^N l_{t,y^N})^{1-\tau_t} \ln(y_0^N l_{t,y^N}). \quad (71)$$

To ensure the aggregation preserves the structure of the planner's intertemporal and intratemporal optimality conditions, we also define coefficients that link the group-level Euler equations to their individual counterparts. For the consumption Euler equation, we introduce $\xi_{y^N}^{u,E}$ such that:

$$\xi_{y^N}^{u,E} u'(c_{t,y^N}) = \beta \mathbb{E}_t \left[R_{t+1} \sum_{\tilde{y}^N \in \mathcal{Y}^N} \Pi_{t+1, y^N \tilde{y}^N} \xi_{\tilde{y}^N}^{u,E} u'(c_{t+1, \tilde{y}^N}) \right] + \nu_{t,y^N}. \quad (72)$$

This condition ensures that aggregate marginal utilities satisfy a properly defined truncated-history Euler equation.

Similarly, for the intratemporal labor-leisure tradeoff, we impose:

$$\xi_{y^N}^{v,1} v'(l_{t,y^N}) = (1 - \tau_t) w_t \xi_{y^N}^y (y_0^N l_{t,y^N})^{1-\tau_t} \xi_{y^N}^{u,1} \frac{u'(c_{t,y^N})}{l_{t,y^N}}. \quad (73)$$

Finally, for the aggregation of expressions that involve second-order derivatives of utility, such as $u''(c_{i,t})$ multiplied by linear combinations of Lagrange multipliers, we introduce coefficients $\{\xi_{y^N}^{u,2}, \xi_{y^N}^{u\ell,2}\}$ such that group-level FOCs can be expressed in terms of $u''(c_{t,y^N})$ and $u'(c_{t,y^N})/l_{t,y^N}$, maintaining the same structural form as the individual FOCs of the planner.

Aggregate technology and government constraints. Aggregate production is $Y_t = K_{t-1}^\alpha L_t^{1-\alpha}$ with $L_t = \sum_{y^N} S_{t,y^N} y_0^N l_{t,y^N}$ and $K_{t-1} = \int_i a_{i,t-1} di - B_{t-1} = \sum_{y^N} S_{t,y^N} \tilde{a}_{t,y^N} - B_{t-1}$.

The (post-tax) household budget identity aggregated by truncated group is:

$$c_{t,y^N} + \tilde{a}_{t,y^N} = R_t \tilde{a}_{t,y^N} + w_t \xi_{y^N}^y \left(y_0^N l_{t,y^N} \right)^{1-\tau_t}, \quad (74)$$

and government resource feasibility is the same as in the main text (with nominal terms and Rotemberg adjustment), using the aggregate Y_t , B_t , and G_t . Market clearing for assets and labor is:

$$K_{t-1} + B_{t-1} = \sum_{y^N} S_{t-1,y^N} \tilde{a}_{t-1,y^N}, \quad L_t = \sum_{y^N} S_{t,y^N} y_0^N l_{t,y^N}. \quad (75)$$

Steady state and identification of the ξ 's. In steady state we compute policy functions and the invariant wealth distribution of the underlying Bewley economy, then aggregate within each y^N -group to obtain $(c_{y^N}, l_{y^N}, \tilde{a}_{y^N})$. The coefficients $\{\xi^{u,0}, \xi^{v,0}, \xi^{u,1}, \xi^{v,1}, \xi^y, \xi^{y \ln}, \xi^{u,\ell}, \xi^{u,2}, \xi^{ul,2}\}$ are pinned down by identities (68)–(71) (and their u'' analogs) evaluated at steady state, ensuring exact aggregation of the nonlinear terms that enter the planner's FOCs.

Aggregate shocks and fixed-coefficient approximation. We follow the standard truncation assumption:

Assumption A. (i) The aggregation coefficients $\{\xi\}$ are held fixed at their steady-state values when solving dynamics with small aggregate shocks. (ii) The set of credit-constrained truncated histories is time-invariant.

These assumptions ensure that the steady state of the truncated model coincides with that of the full model, and that truncated allocations converge to the full equilibrium as N increases.

C.2 Ramsey program in the truncated economy

Objective and constraints. Let ω_{y^N} denote the Pareto weight attached to truncated history y^N . The planner's objective in the truncated economy is:

$$W_0 = \mathbb{E}_0 \sum_{t \geq 0} \beta^t \sum_{y^N \in \mathcal{Y}^N} S_{t,y^N} \left(\omega_{y^N} \left[\xi_{y^N}^{u,0} u(c_{t,y^N}) - \xi_{y^N}^{v,0} v(l_{t,y^N}) \right] \right), \quad (76)$$

subject to: (i) the evolution of population shares (65); (ii) the definition of beginning-of-period wealth by truncated group (67); (iii) aggregate Euler equations (72) and (73); (iv) the group budget identities (74); (v) market clearing (75); (vi) the aggregate production function; (vii) the nominal Phillips block and related definitions from the main text (including M_t and ξ_t); and (viii) the government resource constraint (with Rotemberg term);

The planner chooses $\{\Pi_t, \tilde{R}_t^n, \tilde{r}_t^k, B_t, r_t, w_t, \tau_t\}$ and group allocations $\{c_{t,y^N}, l_{t,y^N}, \tilde{a}_{t,y^N}\}_{y^N}$.

Aggregated multipliers and social value of liquidity. Define the (truncated) social value of liquidity for group y^N as:

$$\begin{aligned} \psi_{t,y^N} &:= \omega_{y^N} \xi_{y^N}^{u,0} u'(c_{t,y^N}) \\ &\quad - \left[\lambda_{c,t,y^N} \xi_{y^N}^{u,E} - (1+r_t) \tilde{\lambda}_{c,t,y^N} \xi_{y^N}^{u,E} - \lambda_{l,t,y^N} \xi_{y^N}^y (1-\tau_t) w_t (y_0^N)^{1-\tau_t} l_{t,y^N}^{-\tau_t} \xi_{y^N}^{u,1} \right. \\ &\quad \left. + \left((\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) - \frac{\epsilon-1}{\psi} \gamma_t (\xi_t - 1) \right) Y_t \right] u''(c_{t,y^N}), \end{aligned} \quad (77)$$

and the net value $\hat{\psi}_{t,y^N} := \psi_{t,y^N} - \mu_t$, where μ_t is the multiplier on the aggregate resource constraint.

FOCs of the truncated Ramsey problem. The FOCs in the truncated model mirror the structure of (53)–(64) with group-level aggregation coefficients. We list them below.

Inflation Π_t .

$$0 = \mu_t \psi (\Pi_t - 1) + (\gamma_t - \gamma_{t-1}) (2\Pi_t - 1) M_t - \left(\Gamma_t (1 - \tau_t^k) B_{t-1} - \beta^{-1} \Psi_{t-1} \right) \frac{\tilde{R}_{t-1}^n}{\Pi_t^2 Y_t}. \quad (78)$$

Nominal bond return $\tilde{R}_t^n$.

$$0 = -\beta \Gamma_{t+1} (1 - \tau_{t+1}^k) \frac{B_t}{\Pi_{t+1}} + \Psi_t \frac{1}{\Pi_{t+1}}. \quad (79)$$

Capital return $\tilde{r}_t^k$.

$$\beta^{-1} \Psi_{t-1} = \left(\left(\frac{\epsilon-1}{\psi} \right) \gamma_t \frac{1}{\alpha} M_t - \Gamma_t (1 - \tau_t^k) \right) K_{t-1}. \quad (80)$$

Group assets $\tilde{a}_{t,y^N}$. Using (90), the net social value of liquidity satisfies:

$$\begin{aligned} \hat{\psi}_{t,y^N} &= \beta \mathbb{E}_t \left[(1+r_{t+1}) \sum_{\tilde{y}^N} \Pi_{y^N \tilde{y}^N} \hat{\psi}_{t+1, \tilde{y}^N} \right] + \beta \mathbb{E}_t \left[\mu_{t+1} \left(\left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \frac{\tilde{r}_{t+1}^k + \delta}{\xi_{t+1}} - r_{t+1} - \delta \right) \right] \\ &\quad + \left(\frac{\epsilon-1}{\psi} \right) \beta \mathbb{E}_t \left[\gamma_{t+1} (\xi_{t+1} - \alpha) M_{t+1} \frac{Y_{t+1}}{K_t} \right] - \beta \mathbb{E}_t \left[(\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \alpha \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ &\quad - \beta \mathbb{E}_t \left[\Gamma_{t+1} (1 - \tau_{t+1}^k) \tilde{r}_{t+1}^k \right] + \beta \mathbb{E}_t [\Gamma_{t+1} r_{t+1}]. \end{aligned} \quad (81)$$

Public debt B_t .

$$\begin{aligned}\mu_t = & \beta \mathbb{E}_t \left[\mu_{t+1} \left(1 - \delta + \left(1 - \frac{\psi}{2} (\Pi_{t+1} - 1)^2 \right) \frac{\tilde{r}_{t+1}^k + \delta}{\xi_{t+1}} \right) \right] \\ & + \beta \mathbb{E}_t \left[\left(\left(\frac{\epsilon-1}{\psi} \right) \gamma_{t+1} (\xi_{t+1} - \alpha) - \alpha (\gamma_{t+1} - \gamma_t) \Pi_{t+1} (\Pi_{t+1} - 1) \right) \frac{M_{t+1} Y_{t+1}}{K_t} \right] \\ & + \beta \mathbb{E}_t \left[\Gamma_{t+1} (1 - \tau_{t+1}^k) \left(\frac{\tilde{R}_t^n}{\Pi_{t+1}} - 1 - \tilde{r}_{t+1}^k \right) \right].\end{aligned}\quad (82)$$

Labor l_{t,y^N} . With the coefficients in (69)–(70),

$$\begin{aligned}\frac{\omega_{y^N} \xi_{y^N}^{v,0} v'(l_{t,y^N}) + \lambda_{l,t,y^N} \xi_{y^N}^{v,1} v''(l_{t,y^N})}{(1 - \tau_t) w_t \xi_{y^N}^y (y_0^N)^{1-\tau_t} l_{t,y^N}^{-\tau_t}} = & \hat{\psi}_{t,y^N} - \lambda_{l,t,y^N} \tau_t \xi_{y^N}^{u,\ell} \frac{u'(c_{t,y^N})}{l_{t,y^N}} \\ & + \mu_t \frac{\left(1 - \frac{\psi}{2} (\Pi_t - 1)^2 \right) F_{L,t}}{(1 - \tau_t) w_t \xi_{y^N}^y (y_0^N)^{-\tau_t} l_{t,y^N}^{-\tau_t}} \\ & + \frac{\left(\left(\frac{\epsilon-1}{\psi} \right) \gamma_t (\xi_t - (1 - \alpha)) - (1 - \alpha) (\gamma_t - \gamma_{t-1}) \Pi_t (\Pi_t - 1) \right) Y_t M_t / L_t}{(1 - \tau_t) w_t \xi_{y^N}^y (y_0^N)^{-\tau_t} l_{t,y^N}^{-\tau_t}}.\end{aligned}\quad (83)$$

Wage w_t .

$$\begin{aligned}0 = & \sum_{y^N} S_{t,y^N} \xi_{y^N}^y (y_0^N l_{t,y^N})^{1-\tau_t} \left(\hat{\psi}_{t,y^N} + \lambda_{l,t,y^N} (1 - \tau_t) \xi_{y^N}^{u,\ell} \frac{u'(c_{t,y^N})}{l_{t,y^N}} \right) \\ & + \left(\frac{\epsilon-1}{\psi} \right) \gamma_t M_t \frac{1}{1 - \tau_t} \left(\frac{w_t}{\kappa} \right)^{\frac{\tau_t}{1-\tau_t}} \frac{1}{\kappa} \frac{L_t}{1 - \alpha}.\end{aligned}\quad (84)$$

Safe return r_t .

$$0 = \sum_{y^N} S_{t,y^N} \left(\hat{\psi}_{t,y^N} \tilde{a}_{t,y^N} + \lambda_{c,t-1,y^N} \xi_{y^N}^{u,1} u'(c_{t,y^N}) \right) + \Gamma_t (B_{t-1} + K_{t-1}).\quad (85)$$

Progressivity τ_t .

$$\begin{aligned}0 = & \sum_{y^N} S_{t,y^N} \xi_{y^N}^y (y_0^N l_{t,y^N})^{1-\tau_t} \left(\hat{\psi}_{t,y^N} + \lambda_{l,t,y^N} (1 - \tau_t) \xi_{y^N}^{u,\ell} \frac{u'(c_{t,y^N})}{l_{t,y^N}} \right) \ln(y_0^N l_{t,y^N}) \\ & + \sum_{y^N} S_{t,y^N} \lambda_{l,t,y^N} \xi_{y^N}^y (y_0^N l_{t,y^N})^{1-\tau_t} \xi_{y^N}^{u,\ell} \frac{u'(c_{t,y^N})}{l_{t,y^N}} \\ & - \left(\frac{\epsilon-1}{\psi} \right) \gamma_t M_t \frac{1}{(1 - \tau_t)^2} \left(\frac{w_t^{\frac{\tau_t}{1-\tau_t}}}{\kappa^{\frac{1}{1-\tau_t}}} \right) \ln \left(\frac{w_t}{\kappa} \right) \frac{L_t}{1 - \alpha}.\end{aligned}\quad (86)$$

We must furthermore have:

$$\tilde{\lambda}_{c,t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \lambda_{c,t-1,\tilde{y}^N}, \quad (87)$$

$$\tilde{a}_{t,y^N} \geq 0 \text{ and } \tilde{a}_{t,y^N} = \sum_{\tilde{y}^N \in \mathcal{Y}^{\infty N}} \frac{S_{t-1,\tilde{y}^N}}{S_{t,y^N}} \Pi_{t,\tilde{y}^N y^N} \tilde{a}_{t-1,\tilde{y}^N}. \quad (88)$$

Link with the primitive FOCs. Equations (78)–(88) reproduce the structure of the primitive Ramsey FOCs (53)–(64), with individual sums replaced by truncated-history aggregates and with the nonlinear aggregation coefficients $\{\xi\}$ ensuring that (i) the truncated steady state coincides with the full model and (ii) local dynamics under small shocks are well-approximated.

C.3 Computation and identification of Pareto weights in the truncated model

Given a truncation length N and a discrete set $\mathcal{Y}$ of idiosyncratic states, we compute the steady-state allocation of the underlying Bewley economy and recover the coefficients $\{\xi\}$. We then form the truncated system (76)–(86), evaluate it at the observed (or calibrated) steady state, and express the planner’s steady-state FOCs as linear restrictions in the Pareto weights $\{\omega_{y^N}\}_{y^N \in \mathcal{Y}^N}$. Using the identification criterion described in Section 3.3, we select the set of Pareto weights that (i) satisfy those restrictions and (ii) are closest to the utilitarian benchmark.

As discussed in the main text, refined truncation can be used to emphasize histories that occur with higher probability by increasing their effective memory length, which improves accuracy while keeping the number of aggregated states linear in the maximum truncation length. The quality of the approximation improves with N , and the steady-state replication is exact by construction under Assumption A.

C.4 Matrix expression

In this section, we provide closed-form formulas for preference multipliers discussed in Section C.1. First, denote: $\circ$ as the Hadamard product, $\otimes$ the Kronecker product, and $\times$ the usual matrix product. For a vector V , $\text{diag}(V)$ is the diagonal matrix with V on the diagonal.

Truncated histories are indexed by y^N , with N_{tot} total histories. Let $\mathbf{S} = (S_{y^N})_{y^N}$ be the steady-state shares and define the stacked vectors from the Bewley steady state:

$$\mathbf{a}, \mathbf{c}, \mathbf{l}, \boldsymbol{\lambda}_c, \boldsymbol{\lambda}_l, u'(\mathbf{c}), u''(\mathbf{c}), v'(\mathbf{l}), v''(\mathbf{l}).$$

Let $\mathbf{y} = (y_0^N)_{y^N}$ collect current idiosyncratic productivities, $\mathbf{I}$ be the $N_{\text{tot}} \times N_{\text{tot}}$ identity, and $\mathbf{\Pi}$ the transition matrix over truncated histories. As before, $\mathbf{P}$ is diagonal with ones on unconstrained histories.

We also stack the NK objects at steady state:

$$\mathbf{Y}, \mathbf{L}, \mathbf{K}, \mathbf{\Pi}^{(\text{infl})}, \boldsymbol{\gamma}, \boldsymbol{\xi}, \mathbf{M},$$

and define the Rotemberg–markup composite (per history) that appears in the primitive FOCs:

$$\boldsymbol{\Theta} := \left((\boldsymbol{\gamma} - \boldsymbol{\gamma}^-) \circ \mathbf{\Pi}^{(\text{infl})} \circ (\mathbf{\Pi}^{(\text{infl})} - \mathbf{1}) - \frac{\epsilon-1}{\psi} \boldsymbol{\gamma} \circ (\boldsymbol{\xi} - \mathbf{1}) \right) \circ \mathbf{Y},$$

where $\boldsymbol{\gamma}^-$ is the (steady-state) one-step lag of $\boldsymbol{\gamma}$. (At a strict steady state one typically has $\boldsymbol{\gamma} = \boldsymbol{\gamma}^-$ and $\mathbf{\Pi}^{(\text{infl})} = \mathbf{1}$, but we keep $\boldsymbol{\Theta}$ explicit for generality.)

Stacked steady-state relations. The share recursion and aggregated individual budget read:

$$\mathbf{S} = \mathbf{\Pi} \mathbf{S}, \tag{89}$$

$$\mathbf{S} \circ \mathbf{c} + \mathbf{S} \circ \mathbf{a} = (1+r) \mathbf{\Pi} (\mathbf{S} \circ \mathbf{a}) + w \mathbf{S} \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau}, \tag{90}$$

with credit constraints $(\mathbf{I} - \mathbf{P})\mathbf{a} = \mathbf{0}$. Here $\boldsymbol{\xi}^y$ is the (steady-state) nonlinearity coefficient from (70).

Nonlinear aggregation coefficients. Let $\mathbf{D}_x$ denote $\text{diag}(\mathbf{x})$. The truncated Euler-aggregation coefficients are obtained exactly as in the Appendix C.1:

$$\boldsymbol{\xi}^{u,E} = \left[\left(\mathbf{I} - \beta(1+r)\mathbf{\Pi}^\top \right) \mathbf{D}_{u'(c)} \right]^{-1} \boldsymbol{\nu}, \tag{91}$$

$$\boldsymbol{\xi}^{v,1} = (1-\tau) w (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \boldsymbol{\xi}^y \circ \boldsymbol{\xi}^{u,1} \circ u'(c) ./ (\mathbf{l} \circ v'(\mathbf{l})), \tag{92}$$

with $\boldsymbol{\xi}^{u,0}, \boldsymbol{\xi}^{u,1}, \boldsymbol{\xi}^{v,0}, \boldsymbol{\xi}^y$ computed from the definitions in (68)–(70) (and $\boldsymbol{\xi}^{y \ln}$ analogously for the progressivity FOC).

C.4.1 Matrix expressions for the FOCs

Define the sized (share-weighted) variables:

$$\bar{\boldsymbol{\lambda}}_l := \mathbf{S} \circ \boldsymbol{\lambda}_l, \quad \bar{\boldsymbol{\psi}} := \mathbf{S} \circ \hat{\boldsymbol{\psi}}, \quad \bar{\mathbf{\Pi}} := \mathbf{S} \circ \mathbf{\Pi}^\top \circ (1/\mathbf{S}), \quad \bar{\boldsymbol{\omega}} := \mathbf{S} \circ \boldsymbol{\omega}, \quad \bar{\boldsymbol{\lambda}}_c := \mathbf{S} \circ \boldsymbol{\lambda}_c,$$

and shortcuts $\tilde{\boldsymbol{\xi}}^{u,1} := \boldsymbol{\xi}^{u,1} ./ \mathbf{l}$, $\tilde{\boldsymbol{\xi}}^{v,1} := \boldsymbol{\xi}^{v,1} ./ ((1-\tau)w \boldsymbol{\xi}^y \circ \mathbf{y}^{1-\tau} \circ \mathbf{l}^{-\tau})$, $\tilde{\boldsymbol{\xi}}^{v,0} := \boldsymbol{\xi}^{v,0} ./ ((1-\tau)w \boldsymbol{\xi}^y \circ \mathbf{y}^{1-\tau} \circ \mathbf{l}^{-\tau})$. Note that $\mathbf{S} \circ \bar{\boldsymbol{\lambda}}_c = \mathbf{\Pi} \bar{\boldsymbol{\lambda}}_c$.

The share-weighted *net* social value of liquidity (group level) solves:

$$\begin{aligned} \bar{\psi} = & \underbrace{D_{\xi^{u,0} \circ u'(c)}}_{\hat{M}_0} \bar{\omega} - \underbrace{D_{\xi^{u,E} \circ u''(c)}}_{\hat{A}_c} \left(\bar{\lambda}_c - (1+r) \Pi \bar{\lambda}_c \right) \\ & + \underbrace{(1-\tau)w D_{\xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u''(c)}}_{\hat{A}_\ell} \bar{\lambda}_l + \underbrace{D_{\Theta \circ u''(c)}}_{\hat{A}_{\text{NK}}} S - \mu S. \end{aligned} \quad (93)$$

Unconstrained truncated histories satisfy the (grouped) implementability condition implied by (60). At the truncated level this yields an affine recursion:

$$P \bar{\psi} = \beta P \left((1+r) \bar{\Pi} \bar{\psi} + \Upsilon \right), \quad (I - P) \bar{\lambda}_c = 0, \quad (94)$$

where the stacked vector Υ captures the (known, steady-state) additive wedge driven by the Γ - Ψ block and nominal-bond return in (60).⁹

The wage, interest-rate, and progressivity FOCs become (stacked):

$$\left(\xi^y \circ (y \circ l)^{1-\tau} \right)^\top \bar{\psi} = -(1-\tau) \left(\xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u'(c) \right)^\top \bar{\lambda}_l, \quad (95)$$

$$a^\top \bar{\psi} = - \left(\xi^{u,E} \circ u'(c) \right)^\top \Pi \bar{\lambda}_c, \quad (96)$$

$$\left(\ln(y \circ l) \circ \xi^y \circ (y \circ l)^{1-\tau} \right)^\top \bar{\psi} = - \left((1 + (1-\tau) \ln(y \circ l)) \circ \xi^y \circ (y \circ l)^{1-\tau} \circ \tilde{\xi}^{u,1} \circ u'(c) \right)^\top \bar{\lambda}_l. \quad (97)$$

C.4.2 Solving the system

From the labor FOC block, solve for $\bar{\lambda}_l$ as:

$$D_{\tilde{\xi}^{v,1} \circ v''(l) + \tau \tilde{\xi}^{u,1} \circ u'(c)} \bar{\lambda}_l = \bar{\psi} - D_{\tilde{\xi}^{v,0} \circ v'(l)} \bar{\omega} + \mu \Phi, \quad \bar{\lambda}_l = M_0 \bar{\omega} + M_1 \bar{\psi} + \mu V_0, \quad (98)$$

where:

$$\begin{aligned} M_1 &:= \left(D_{\tilde{\xi}^{v,1} \circ v''(l) + \tau \tilde{\xi}^{u,1} \circ u'(c)} \right)^{-1}, \quad M_0 := -M_1 D_{\tilde{\xi}^{v,0} \circ v'(l)}, \\ \Phi &:= F_L S / \left((1-\tau)w \xi^y \circ y^{-\tau} \circ l^{-\tau} \right), \quad V_0 := M_1 \Phi. \end{aligned}$$

Plug (98) into (93) to obtain:

$$\bar{\psi} = \underbrace{\left(\hat{M}_0 + \hat{A}_\ell M_0 \right)}_{N_0} \bar{\omega} + \underbrace{\left(-\hat{A}_c + \hat{A}_\ell M_1 \right)}_{N_1} \bar{\lambda}_c + \underbrace{\left(\hat{A}_{\text{NK}} S - \mu S + \hat{A}_\ell V_0 \mu \right)}_{n_0 + \mu n_1}. \quad (99)$$

⁹Explicitly, $\Upsilon := \mathbb{E}[\Gamma_{t+1}(r_{t+1} - (1 - \tau_{t+1}^k)(\tilde{R}_t^n / \Pi_{t+1} - 1))]$ stacked by truncated history; at the steady state it is evaluated using the truncated transition Π .

Use (94) to eliminate $\bar{\lambda}_c$. Write $\mathbf{Q} := \beta \mathbf{P}(1+r) \bar{\Pi}$ and $\mathbf{q} := \beta \mathbf{P} \Upsilon$. Then:

$$\mathbf{P}\bar{\psi} = \mathbf{Q}\bar{\psi} + \mathbf{q}, \quad (\mathbf{I} - \mathbf{P})\bar{\lambda}_c = \mathbf{0}.$$

Substituting (99) and solving for $\bar{\lambda}_c$ gives:

$$\bar{\lambda}_c = \mathbf{M}_5 \bar{\omega} + \mu \mathbf{V}_2 + \mathbf{V}_\Upsilon, \quad (100)$$

with:

$$\begin{aligned} \mathbf{M}_5 &:= -\left((\mathbf{I} - \mathbf{P}) + \mathbf{P}(\mathbf{I} - \mathbf{Q})\mathbf{N}_1\right)^{-1} \mathbf{P}(\mathbf{I} - \mathbf{Q})\mathbf{N}_0, \\ \mathbf{V}_2 &:= -\left((\mathbf{I} - \mathbf{P}) + \mathbf{P}(\mathbf{I} - \mathbf{Q})\mathbf{N}_1\right)^{-1} \mathbf{P}(\mathbf{I} - \mathbf{Q})\mathbf{n}_1, \quad \mathbf{V}_\Upsilon := -\left((\mathbf{I} - \mathbf{P}) + \mathbf{P}(\mathbf{I} - \mathbf{Q})\mathbf{N}_1\right)^{-1} \mathbf{q}. \end{aligned}$$

Back-substitute (100) into (99):

$$\bar{\psi} = \mathbf{M}_6 \bar{\omega} + \mu \mathbf{V}_1 + \mathbf{V}_\Upsilon^\psi, \quad \bar{\lambda}_l = \hat{\mathbf{M}}_6 \bar{\omega} + \mu \hat{\mathbf{V}}_0 + \hat{\mathbf{V}}_\Upsilon, \quad (101)$$

where:

$$\begin{aligned} \mathbf{M}_6 &:= \mathbf{N}_0 + \mathbf{N}_1 \mathbf{M}_5, \quad \mathbf{V}_1 := \mathbf{n}_1 + \mathbf{N}_1 \mathbf{V}_2, \quad \mathbf{V}_\Upsilon^\psi := \mathbf{n}_0 + \mathbf{N}_1 \mathbf{V}_\Upsilon, \\ \hat{\mathbf{M}}_6 &:= \mathbf{M}_0 + \mathbf{M}_1 \mathbf{M}_6, \quad \hat{\mathbf{V}}_0 := \mathbf{M}_1 \mathbf{V}_1 + \mathbf{V}_0, \quad \hat{\mathbf{V}}_\Upsilon := \mathbf{M}_1 \mathbf{V}_\Upsilon^\psi. \end{aligned}$$

Use (96) to pin down μ :

$$\mu = -\frac{\mathbf{a}^\top (\mathbf{M}_6 \bar{\omega} + \mathbf{V}_\Upsilon^\psi) + (\boldsymbol{\xi}^{u,E} \circ u'(c))^\top \Pi \mathbf{M}_5 \bar{\omega} + (\boldsymbol{\xi}^{u,E} \circ u'(c))^\top \Pi \mathbf{V}_\Upsilon}{\mathbf{a}^\top \mathbf{V}_1 + (\boldsymbol{\xi}^{u,E} \circ u'(c))^\top \Pi \mathbf{V}_2}. \quad (102)$$

Constructing the linear constraints on ω . Insert (101) and (102) into the wage and progressivity FOCs (95)–(97). This yields two linear constraints in $\bar{\omega}$:

$$\tilde{\mathbf{L}}_1 \bar{\omega} = \tilde{\mathbf{b}}_1, \quad (103)$$

$$\tilde{\mathbf{L}}_2 \bar{\omega} = \tilde{\mathbf{b}}_2, \quad (104)$$

with:

$$\begin{aligned} \tilde{\mathbf{L}}_1 &:= (\boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau})^\top \mathbf{M}_6 + (1-\tau) (\boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(c))^\top \hat{\mathbf{M}}_6, \\ \tilde{\mathbf{L}}_2 &:= (\ln(\mathbf{y} \circ \mathbf{l}) \circ \boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau})^\top \mathbf{M}_6 + (1 + (1-\tau) \ln(\mathbf{y} \circ \mathbf{l}))^\top (\boldsymbol{\xi}^y \circ (\mathbf{y} \circ \mathbf{l})^{1-\tau} \circ \tilde{\boldsymbol{\xi}}^{u,1} \circ u'(c))^\top \hat{\mathbf{M}}_6, \end{aligned}$$

and right-hand sides $\tilde{\mathbf{b}}_1, \tilde{\mathbf{b}}_2$ collecting all constants produced by $\boldsymbol{\Theta}, \Upsilon$, and the μ term in (102). (They are fully determined once the steady-state truncated environment is computed.)

Finally, map period-productivity weights $\boldsymbol{\omega}^Y = (\omega_y)_y$ to history weights via the usual

selectors:

$$\mathbf{R}_0 = \begin{bmatrix} \mathbf{1}_{S_{y_1}} & 0 & \cdots & 0 \\ 0 & \mathbf{1}_{S_{y_2}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \mathbf{1}_{S_{y_Y}} \end{bmatrix}, \quad \mathbf{R}_1 := \mathbf{D}_S \mathbf{R}_0, \quad \bar{\boldsymbol{\omega}} = \mathbf{R}_1 \boldsymbol{\omega}^Y.$$

The two linear constraints on $\boldsymbol{\omega}^Y$ are:

$$\mathbf{L}_1 \boldsymbol{\omega}^Y = \mathbf{b}_1, \quad \mathbf{L}_2 \boldsymbol{\omega}^Y = \mathbf{b}_2, \quad \text{with } \mathbf{L}_j := \tilde{\mathbf{L}}_j \mathbf{R}_1, \quad \mathbf{b}_j := \tilde{\mathbf{b}}_j \quad (j = 1, 2). \quad (105)$$

C.4.3 Constrained weights

Let $\boldsymbol{\omega}_U \in \mathbb{R}^Y$ collect the Y productivity-level weights we want to estimate. Let $\mathbf{D}_S := \text{diag}(\{S_{y^N}\}_{y^N})$ be the diagonal matrix of truncated-history sizes, and let $\mathbf{M}_7$ be the $N_{\text{tot}} \times Y$ incidence matrix that maps each truncated history to its current productivity level. The stacked history-level weights are:

$$\bar{\boldsymbol{\omega}} = \mathbf{D}_S \mathbf{M}_7 \boldsymbol{\omega}_U.$$

Observe $\mathbf{L}_1$ and $\mathbf{L}_2$ are two linear constraints obtained from the planner's FOCs in the truncated model. Identification under the NK model then amounts to the following quadratic program:

$$\min_{\boldsymbol{\omega}_U} \mathbf{S}^P \left\| \boldsymbol{\omega}_U - \mathbf{1}_Y \right\|^2 \quad \text{s.t.} \quad \mathbf{L}_1 \mathbf{D}_S \mathbf{M}_7 \boldsymbol{\omega}_U = \mathbf{0}, \quad \mathbf{L}_2 \mathbf{D}_S \mathbf{M}_7 \boldsymbol{\omega}_U = \mathbf{0},$$

together with the normalization:

$$\mathbf{1}_{N_{\text{tot}}}^\top \mathbf{D}_S \mathbf{M}_7 \boldsymbol{\omega}_U = 1,$$

and, when imposing element-wise monotonicity of the SWF, the positivity constraint:

$$\boldsymbol{\omega}_U \geq \mathbf{0}.$$